

Robust parameters determination for ergonomical product design via computer musculoskeletal modeling and multi-objective optimization

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ABSTRACT

This research aims to determine the optimum parameter values for ergonomic product designs via computer musculoskeletal modeling (CMM) and multi-objective optimization (MOO). The multiple-muscle activities measured by the AnyBody (AB) Modeling System are used to develop the functional relationships with product design parameters via a statistical method, such as design of experiment (DOE). One DOE, Response Surface Methodology (RSM), is adopted in the present approach. Such functional relationships are considered as objective functions which will be further formulated via compromise programming (CP) for multi-objective optimization (MOO) problems. A bike-frame design problem is chosen herein as an example to demonstrate the proposed method; it includes determining the lengths of the stem, head tube, fork, top tube, seat post, seat tube and pedal crank. Two cases related to the proposed approach are also introduced: one is a deterministic product design under certain conditions, and the other is a robust product design under uncertain conditions. Because the combination of computer musculoskeletal modeling, statistical method, and multi-objective optimization technique is realized in the proposed approach, ergonomic product designs for safety and efficiency under uncertainty can be achieved in the early stage of other product designs.

1. Introduction

It is well known that a poor ergonomic product design can lead to discomfort and even health disorders (e.g., back pain, neck and shoulder complaints, etc.) entailing major cost to society through missed work/reduced productivity, physiotherapy, and diminished work-effectiveness (Luttman et al., 2003; Major & Vezina, 2015; Niu, 2010). Thus, the ever-increasing demand by customers for products with improved performance and quality is complemented by an equally strong demand for products with improved comfort and health. These demands provide a possible way to differentiate products from those of competitors. Consequently, product design is forced to place additional emphasis on ergonomics for safety and health reasons in order to gain a competitive edge in the market (Azadeh & Sheikhalishahi, 2015).

Over the past decades, Computer-Aided Engineering (CAE) has revolutionized product development (Jeang, 2008; Jeang, Liang, & Chung, 2008, 2011; Pandian, Hassim, & Hurme, 2013). The current available analysis facilities cover almost every conceivable technical property, and enable technical products to be developed via computer simulation before actual production. As CAE plays the main role, it has

clearly become the requisite tool for modern industries. In addition to its contribution to the conventional applications in the manufacturing sector, CAE is being applied more rigorously to address the issue of product design for safety and health considerations. That is, the use of computer models for effectively integrating humans and products through their analyses highlights the process of product development; meanwhile, attention is also being paid to the ergonomic methods of product development (Damsgaard et al., 2006; Ma et al., 2009; Gragg, Cloutier, & Yang, 2013). Ideally, in the early stages of product design, as aforementioned, a new product can be examined for its degrees of comfort and health by carrying out computer simulations of human interactions with the introduced product. To enhance a product's competitiveness, computer simulation of the mechanics of the human body (the musculoskeletal system) has become a subject of importance in the essential research on the many relevant possible applications.

However, before these computer simulations become efficacious comfort and health assessment tools, the important problems of identifying comfort and health measures and establishing their relations to product design variables have to be solved. In taking product design bioinstrumentation measurements, the electromyogram (EMG),

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electroencephalogram (EEG), and electrocardiogram (ECG) are often employed to evaluate products ergonomically and quantitatively (Chaffin, Andersson, & Martin, 1999; Tsang & Vidulich, 2006; Petrone et al., 2015). Surface EMG is frequently used for evaluating physical stress and fatigue because of its noninvasive nature and ease of measurement (U.S. Department of Health & Human Services, 1992).

Currently, there are numerous commercialized human simulation tools available for job design and posture analysis, such as 3DSSPP (Chaffin et al., 1997), Jack (Badler, Phillips, & Webber, 1993), VSR (Yang, Marler, Kim, & Arora, 2004), AnyBody (Damsgaard et al., 2006; The AnyBody™ Modeling System, Version 4.2.0). From the viewpoint of analysis, particularly in support of dynamic full body modeling, which involves quite a few muscles, the AnyBody Modeling System (AB) can be built for handling models that are substantially more intricate than other CAE technologies could handle. Additionally, an appropriate human simulation tool which can grasp the interaction between design parameters and the human body is essential to successfully perform ergonomic product designs. Until now, there has been no quantitative method like AB that is capable of effectively and efficiently analyzing the ergonomic quality of products. Furthermore, the continuous quantitative measurement score generated from AB makes the design of experiment (DOE) statistical method easily implementable. When the DOE method is implemented to provide the objective functions, optimization techniques such as MOO become usable. The above facts comprise the optimal solutions to be precisely and effectively obtained in the presented research. Thus, the AB Modeling System was adopted in this research. With the aid of AB, the body is perceived as a real multi-body mechanical system connected by joints and actuators in the form of a mechanism.

The usage of certain products involves complex and rhythmic activities, requiring the well-integrated efforts of brain, nerves, skeletal structure, and muscles (Cesar, Javier, & Ricardo, 2014). The coordinated action of various muscles is required to generate the synchronized forces needed to control their performance, so that the result is harmonious and efficient. The inherent assumption about rational coordination is that the CNS will try to smoothly and efficiently balance all the involved muscles in an optimal manner in order to minimize the total muscle activity, so that fatigue and injury can be reduced as much as possible. Because of the rationality assumption, the MOO technique will be introduced to formulate the multi-muscle activities for optimizing the product parameters design in order to achieve optimal health and efficiency. Because MOO is generally formulated with a set of objective functions for the study of related problems, a way to provide these multiple objective functions in advance becomes critically important in the presented approach.

A set of objective functions is unknown most of the time, particularly regarding the functions describing the dynamic interaction between the product design variables and the human musculoskeletal system under uncertain environments. One possible approach is to run physical experiments in building these functions. As is known, physical experiments are costly, inefficient, difficult, and impossible in some cases, so replacing physical experiments with computer simulation, as in the aforementioned discussion, is one of the best methods available for obtaining experimental data in most current engineering applications. These experimental data are then further analyzed by a statistical method to obtain the response functions used for formulating MOO. Thus, an integration of computer simulation and a suitable statistical method becomes crucial for achieving successful applications in the proposed approach (Jeang, 2008; Jeang, Liang, & Chung, 2008). With physical stress and fatigue being important elements of concern in this study, the muscle activity measurements generated from the AB computer software are considered as the computer's experimental data in the proposed approach. These experimental data will then be analyzed by a DOE statistical method, such as response surface methodology (RSM), to find the desired response functions for MOO (Myers, 1991; Myers, Montgomery, & Anderson-Cook, 2009).

In order to improve the ergonomics of product design and refine workplace tasks, the interaction between humans and product design has to be considered early and thoroughly in the lifecycle of products. Such interaction can be described by the known human body posture and the parameter values in product design. As aforementioned, posture can be measured by CAE, such as AB, which is capable of analyzing the musculoskeletal system of humans. Determining how to use the AB Modeling System to carry out an ergonomic product design efficiently becomes our concern. Without mentioning the disadvantages of adopting other CAE in the posture analysis, the conventional approach is performed by an exhaustive strategy, such as the trial and error method to create a set of alternative product designs. Subsequently, the related posture analyses through AB are carried out to evaluate these alternative designs correspondingly. Finally, the alternative design with the finest ergonomic evaluation score is selected from the creation alternatives, with the hope that the parameter values associated with the selected design candidate truly represent the optimal parameter decision for a product design; however, this strategy is very inaccurate, ineffective, and costly. In addition, product design is an evolutionary process requiring constant revision and modification, which usually entails a lot of sensitivity analyses over a period of time. Because of the aforementioned disadvantages, the existing methods prove very tedious, expensive, and time consuming in seeking the best candidates arbitrarily (and often in vain) to determine the truly optimal ergonomic product parameter values. In particular, the degrees of the above shortcomings become even worse for a product designed under an uncertain environment. For example, uncertainties arise from human body elements, such as sizes and weights; thus, the proposed approach adopts the DOE method, RSM, to efficiently create representative alternative designs and construct precise functional relationships between ergonomic product parameter values and ergonomic evaluation scores. With the aid of optimization techniques, these functional relationships can be promoted to formulate the design problem in relation to MOO to ensure that the determined product parameter values are the most representative and accurate optimal solutions. Owing to the reusable features of functional relationships, it turns out to be very convenient in re-conducting MOO for repeatedly revising product designs during the evolutionary design process. Based on the aforementioned discussions, the contributions of the proposed approach, which reflect the deficiencies in existing methods, are listed in the following six outlines.

1. AB scores represent the equivalent EMG measurements which reflect the level of fatigue and the risk of injury for the combined consequences from the human body and the given product design.
2. By applying CAE, AB software, the ergonomic design can be developed early, before the product is actualized.
3. Due to the integration of the computer software AB system and statistical method RSM, ergonomic design functions (the functional relationships between the ergonomic evaluation scores and the product design parameters) can be constructed for design optimization. In addition, after the RSM analysis, the obtained ANOVA is capable of providing the suggestions for continuous product revision.
4. MOO can be formulated from the determined ergonomic design functions. Because MOO is formulated via Mathematical Programming (MP), the derived ergonomic design functions can also be devised as constraints in design problem formulation for design sensitivity analysis. The design sensitivity analysis enables the exploration of the feasible design space, which provides critical information for product revision, particularly under uncertain environments or various applications. For example, by taking advantages of the MP formulation, a sensitivity analysis can be performed by varying the percentage upper levels for adapting to any particular working environment. Consequently, a customized product design for further safety and health improvement becomes

possible.

5. The obtained ergonomic design functions are reusable for further design improvement, and can also be integrated within other fields for advanced product design. The example here in employs ANSYS software for simultaneous ergonomic and mechanical designs (ANSYS Manual, 1997; Jeang, Liang, & Chung, 2008).
6. The random case introduced in the example demonstrates the possibility of product design under uncertain environments. For example, the uncertain human weight is plugged into AB software to generate evaluation scores for a robust product design under uncertainty.
7. With the introduction of the proposed approach, musculoskeletal disorders (MSDs) can be avoided during the entire product lifecycle application. In addition to the cost reduction in product development, the cost savings to society are significant because of the reduction in physiological disorders.

This paper is divided as follows: Sections 2,3, and 4 contain relevant information regarding the presented approach. The information includes computer musculoskeletal modeling, optimization under uncertainty, computer experimental data examined through RSM for response functions, and normalized response functions for formulating the multi-objective optimization of multiple muscle activities. Section 5 introduces block diagrams for the proposed approach. Section 6 provides an application to demonstrate the proposed approach. A relevant discussion is offered in Section 7 for illuminating the proposed approach. Finally, the conclusion is given in Section 8.

2. Response surface method (RSM)

The relationship between the dependent variables and independent variables is usually extremely complex or unknown; however, RSM provides a procedure for solving this problem (Myers, 1991; Myers et al., 2009). Assume that the designer is concerned with a system involving dependent variable Y (responses or ergonomic evaluation scores or muscle activity measurement scores, etc.), which depends on the independent variables (controllable variables or design variables, product parameters, etc.) X_i , which need to be decided so that the best outcomes, Y , can be obtained. In this study, Y signifies muscle activity measurements and X_i is the product parameter values. It is also assumed that X_i is continuous and controllable. For example, in the proposed approach, various levels of product parameter values are combined according to the RSM experimental design matrix for each trial. The combined levels are entered into the computer software package, AB, to derive the corresponding response values, such as muscle activity measurements. A statistical regression analysis under RSM, with a set of muscle activity measurements versus combined product parameter levels, is then performed to find the response functions which are the functional relationships (ergonomic design functions or product design functions) between the dependent variable (responses or muscle activity measurements) Y and n independent variables (controllable variables or design variables), $X_1, X_2, \dots, X_n$. The eventual goal of RSM is to effectively determine the optimal product design values and response functions statistically in the system. Based on the results, ANOVA from RSM, the design engineer can select the critical controllable design variables to significantly reduce the variation of the quality values. Because of the just found optimal values being based on a single objective function, the found response functions are then further applied for MOO to determine the product parameter values for ergonomic designs.

3. The problem in optimizing analytical functions or running CAE simulation under uncertainty

The inherent assumption in finding the optimal objective values through known analytical functions (conversion process) involves the

average value of the controllable variable and the fixed conditions with the absence of uncontrollable variables (or noise impacts) during the optimization process. With this assumption, the corresponding optimal objective values are found based on the determined optimal average values of the controllable variables. However, the preceding optimal objective values only represent one possible response from the design spaces, especially under uncertain environments. Without mentioning the additional uncertainty arising from given conditions and noise impacts, the uncertainty affecting the controllable variables will cause the optimal objective values found directly from analytical functions to be less representative of feasible design spaces. Consequently, deviation from the truly representative objective values will result. The deviation becomes worse when the analytical functions are in nonlinear form. Because the variation of the response value Y always exists, it is increased with the greater variance of the controllable variable X . Other than the fact that the variation of Y results from the variation of the design values of X , uncertain impacts may arise as well from the uncertain condition and noise. The enlarged variation of objective values may jeopardize the stability and validity of decision making in determining the optimal values of the controllable variables. Thus, the variation of objective values caused by the variation of controllable variables has to be considered.

In regard to considering uncertainty in the design decisions, it is important that the objective values must be the measurement scores which best represent all the possible design alternatives. The most representative measurements are then used to construct the functional relationships for further application in finding the optimal values of the controllable variables. These optimal values must signify the best robust decision in an uncertain environment. In this regard, one possible way is to construct functional relationships (response functions or design functions) via the subsequent block diagram depicted in Fig. 1. With the constructed functions, the optimal values of constructed functions and controllable variables are then determined through optimization techniques. However, in some cases, the above analytical functions to provide the response values in finding the functional relationships may not be known in advance. Then, as indicated in the diagram in Fig. 1, the developed CAE computer software can be adopted to create responses of interest for constructing the functional relationships. These functions can then be formulated for optimization to determine the optimal values of controllable variables. In the proposed approach, because the analytical design functions are unknown, the AB software is used to generate response values, muscle activity measurements, for constructing the functional relationship.

4. Multiple objective optimization for multiple muscle activities

Muscles are activated by the central nervous system (CNS) via a complicated electrochemical process (Cesar et al., 2014). The number of muscles available is generally larger than strictly needed to drive most motions; this is often referred to as the redundancy problem of muscle recruitment. AB solves the redundancy problem by means of the min/max criterion. Based on this criterion, all muscles are distributed and coordinated to balance the external load, which makes the maximum relative load of each muscle in the system as small as possible. Because the use of single-joint equilibrium to estimate muscle forces and joint loads in a multi-joint system becomes unjustified in some cases, multi-objective optimization is needed for musculoskeletal modeling during the practical exercise.

In this study, the response functions from RSM represent different muscle activity measurements. Compromise Programming (CP) is used as the means for MOO in the proposed approach. The general formulation of CP is:

$$\text{Min } F(x) = \left\{ \sum_{e=1}^n \alpha_e^q \left| \frac{f_e - f_e^*}{f_{e,\max} - f_e^*} \right|^q \right\}^{1/q} \quad (1)$$

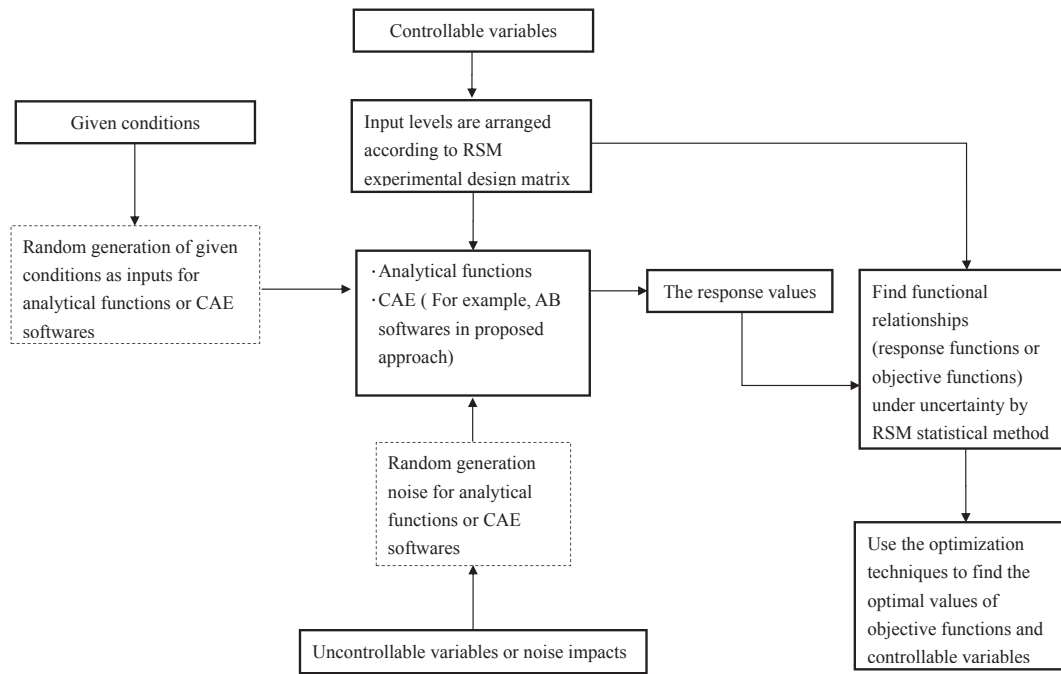


Fig. 1. Block diagram for generating response values with controllable variables, given conditions and uncontrollable variables.

$$s. t. \quad g(x) \leq 0 \quad (2)$$

$$h(x) \leq 0 \quad (3)$$

f_e^* is the optimal value of f_e , $f_{e,\max}$ is the worst value, which is obtained by comparison among all the objective values of all functions, f_e , at the optimal values of controllable variables of f_e^* ; α_e is the weight corresponding to the individual objective function; q is ranged between 1 and ∞ . The large values of q represent the resulting great penalty because of the deviations between the found optimal solutions and ideal optimal solutions. Thus, a large value of q will force the found optimal solutions closer to the ideal optimal solutions.

5. The proposed approach

In the musculoskeletal model, many muscles share the same activity level and contribute to the overall load corresponding to their individual strengths, referred to as “the muscle activity envelope.” This envelope serves to encapsulate the combined load on the muscles. Considering that muscle activity represents the maximum muscle activation in the model, muscle activity can be interpreted as the total percentage of maximum muscle force. A given muscle is loaded to its maximum strength when its activity is 100%. The muscle naturally feels more fatigue with higher muscle activity. In addition to fatigue, the possibility of injury increases. AB is capable of analyzing the human musculoskeletal system as a rigid body; this software package (The AnyBody™ Modeling System, Version 4.2.0) includes a way to solve the multi-body inverse dynamics problem. In this study, the percentage of maximum muscle force from AB is used to represent the level of muscle activity applied for computer musculoskeletal modeling in the evaluation and comparison of a product design. Thus, AB software is considered as the requisite tool to generate the computer experimental data for statistical analysis in seeking to find the response functions further used for MOO in the approach proposed herein.

To carry out the experimental design methods, a set of experimental data for statistical analysis is needed. With level combinations suggested by the RSM experimental design matrix being input into the AB software, the corresponding computer experimental data are generated for each trial. These data are the muscle activity measurements ready for the next statistical analysis; the aim is to obtain the response

functions. These response functions also represent the measurements of the muscle activity functions. Since the optimal solutions found via RSM is the statistical optimization with respect to a single muscle of interest (or single response function, or single objective function), an optimization technique, such as mathematical programming, is needed to integrally formulate multiple muscles of interest into one problem. Additionally, for ensuring fair evaluation, the response functions representing the various muscles of interest need to be normalized by compromise programming prior to the formulation for MOO. Hence, cooperation among the computer musculoskeletal system (AB computer software), statistical method, RSM and mathematical programming enables the statistical and multiple optimization of product parameter determination to become ergonomically workable before the customer's practical exercise. Based on the preceding description, a block diagram representing the proposed approach is shown in Fig. 2.

6. An application

Over a period of time spent in studying the mechanics of bicycle frames with the goal of ergonomically optimizing performance and comfort, it becomes clear that the design problem requires riders' musculoskeletal systems and bicycle frames to be simultaneously modeled in one problem (Oakman & Chan, 2015). Thus, an approach linking the considerations, such as the positions of arms and torso for different physiques and bicycle frames has to be introduced at the same time. An optimal saddle position enables the rider to reduce the muscular effort required for any given power produced. It is known that a good bike position includes three elements: saddle height, saddle-to-bar distance, and fore/aft saddle position. Thus, determining the correct saddle position is of great importance in finding the suitable fit for a road cyclist and the bicycle frame.

As is known, knee injury accounts for 41.7% of bicycle injuries among road cyclists (Schwellnus & Derman, 2005), a fact which may be strongly associated with the unsuitability of the bicycle frame (Asplund & St. Pierre, 2004; Holmes, Donkin, & Witten, 1994). Holmes et al. (1994) indicate that the consequence of a very low or very forward seat in relation to handlebars will place great stress on the patella and femur during the stepping; this process usually results in a narrower space between these two bones. The narrower space continues to increase the

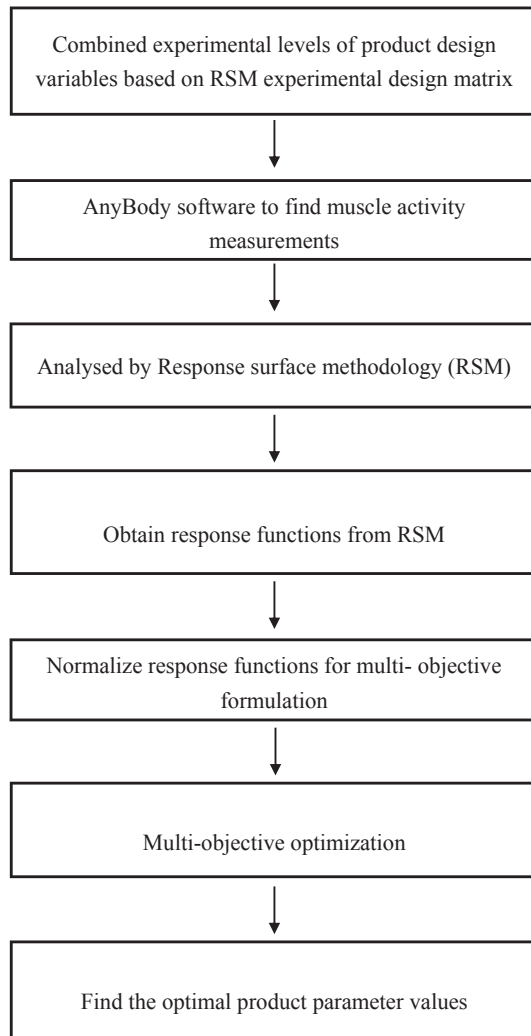


Fig. 2. A block diagram for proposed approach.

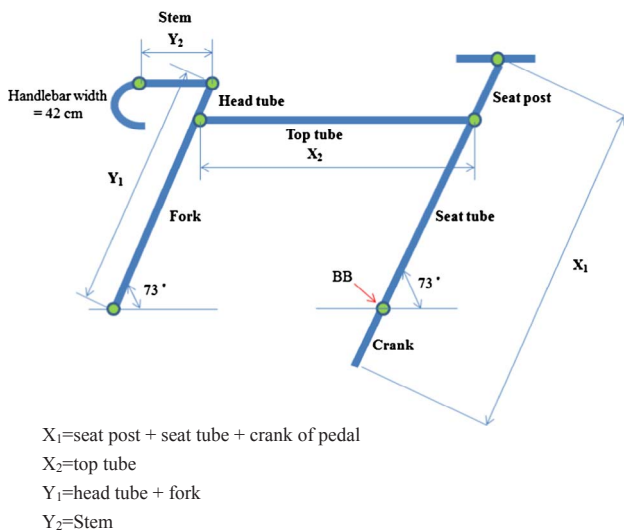


Fig. 3. A bike-frame with product design variables X_1 , X_2 , Y_1 , and Y_2 .

friction between the patella and femur. The high friction will eventually lead to patellar softening. If the seat is too high or too backward, the stepping is likely to cause inflammation of the biceps femoris because of its excessive extension. In this regard, a given bike frame with product

design variables (controllable variables) X_1 , X_2 , Y_1 , and Y_2 , as defined in Fig. 3, most related to the road cyclists' comfort and health, is used as an example to determine the associated product parameter values in order to demonstrate the proposed approach.

Because the proposed approach adopts computer musculoskeletal-related AB software, the weights and heights are essential elements which need to be provided before running this software. As with the theme introduced in Fig. 1 and the flow indicated in Fig. 4, two ways to generate the pair inputs, namely weights and heights, are introduced for random and nonrandom cases, respectively. The nonrandom case is applied mainly for a specific customer design. Conversely, the random case is not used for a particular application; instead, the random case is for public application. For the nonrandom case in Section 6.1, the mean values of weight and height, 69.6 kg and 168 cm, are assigned and input for AB. For the random case, Section 6.2, the input weight and height are generated from the normal distributions, $N(69.6, 3^2)$ and $N(168, 2^2)$, before running AB. Then, through AB, with generated inputs and given conditions, 90 rpm (0.6667 sec/cycle), muscle activity measurements were obtained for the Biceps Femoris Caput Breve, Biceps Femoris Caput Longum, Gastrocnemius, Rectus Femoris, Tibialis Anterior, Vastus Lateralis and Vastus Medialis. These muscles are shown in Fig. 5 and defined in Table 1. They were chosen because they are the most representative of the main muscular groups which are active in cycling.

In order to effectively perform the experimental runs suggested by RSM, three levels of product design parameters are assigned in Table 2, and the experimental design matrix which is introduced in advance is shown in Table 11 of Appendix A. The symbolic -1 , 0 , and $+1$ in Table 11 represent the low level, medium level, and high level, respectively, which are defined in Table 2. Clearly, there are 27 runs or trials in the presented application. With Tables 2 and 11, various level combinations of product parameters are run through AB software under given conditions. Seven muscle activity measurements R_1 , R_2 , R_3 , R_4 , R_5 , R_6 , and R_7 are then simultaneously observed for the corresponding groups of muscles. The rationality illustrated in Table 3 emphasizes that an identical level combination for each run or trial can generate seven response values (muscle activity measurements) simultaneously. For the example of the nonrandom case, under the identical level arrangement, as given in the first run of Table 11, the corresponding plots of muscle activities for R_1 – R_7 are depicted in Fig. 6. The functional relationships between the muscle activity measurements and product design variables X_1 , X_2 , Y_1 , and Y_2 for respective muscles are constructed by RSM statistical analysis. The statistical method, RSM, can be easily executed by computer software SAS to analyze data statistically (Barr & Goodnight, 2008). There are seven functional relationships (muscle activity measurement functions or response functions), R_1 – R_7 . After these functions are normalized, they are further used to formulate the next MOO problem.

6.1. For nonrandomized height and weight of cyclists' case

The analyses of variance (ANOVA) for muscle activities are illustrated in Tables 12–18 of Appendix B. By referring to these tables, the F-ratios and F-rankings for R_1 – R_7 were chosen for summarizing in Table 19 of Appendix B. The combined F-rankings in the last column of Table 19 clearly reveal that the important order of the product design variables is X_1 , X_2 , Y_1 , and Y_2 . Based on this order, the critical design variable is X_1 . The explanation is that the two-joint gastrocnemius muscle plays a greater role in relation to the knee-joint angle in determining muscle activation. The combined effect of the movement at the ankle and knee joints results in an increased length of the gastrocnemius at the higher seat-height position in contrast to the shortened length of gastrocnemius at the lower seat-height position. Clearly, seat height plays a role in the activity patterns of the gastrocnemius. Thus, it is confirmed that setting a proper seat height is the most important of these four design variables (Sanderson & Amoroso, 2009). If further improvement of the ergonomic product design is needed, these critical

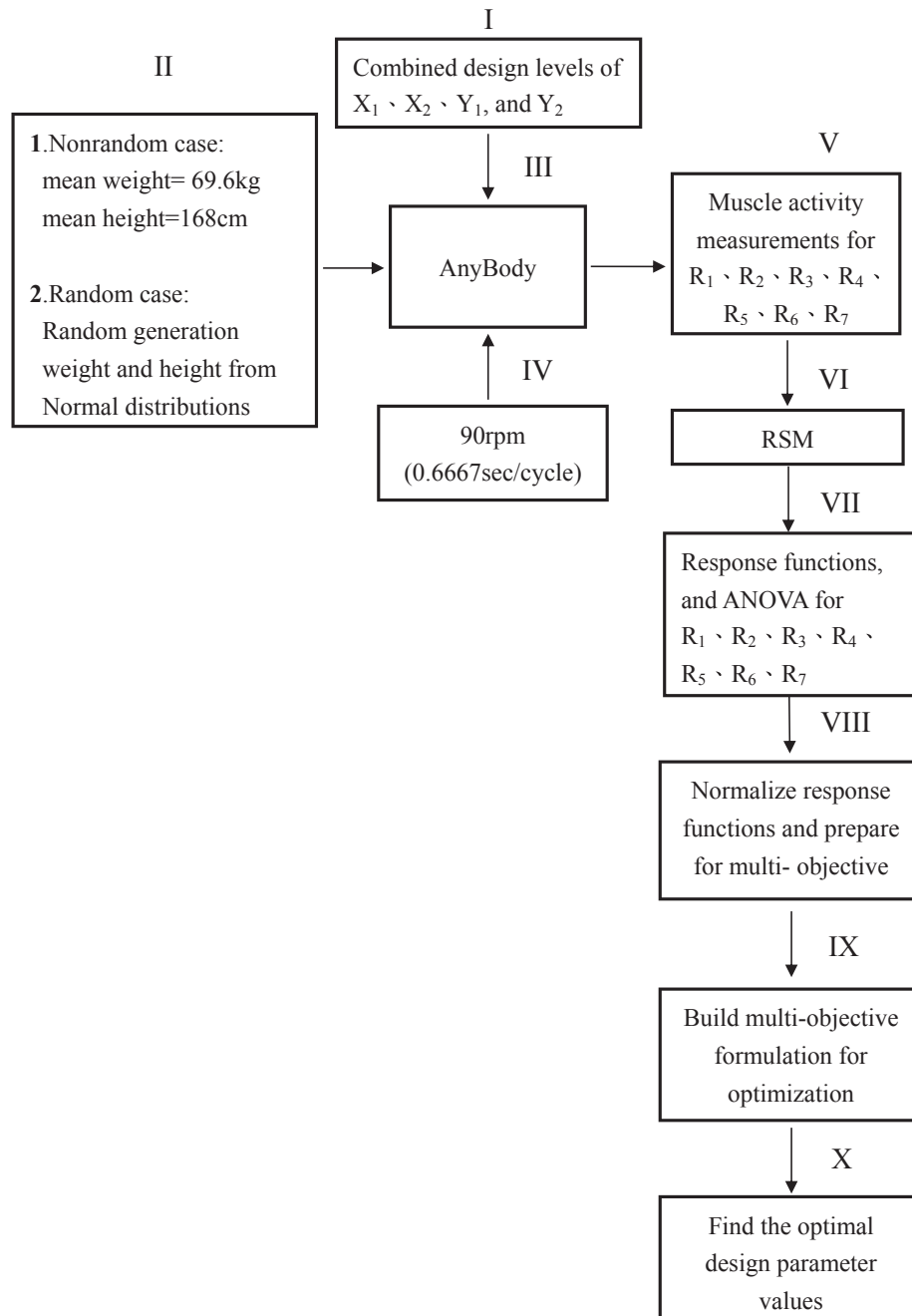


Fig. 4. Block diagram representing the flow of product design parameter determination for nonrandom and random cases.

design variables should be closely monitored and be given high priority for enhancement. After the RSM regression analyses, the values, R^2 , and response functions, R_1 – R_7 , were listed in Table 20. The values, R^2 , are all greater than 0.50 for R_1 – R_7 , which signifies that the second order models provide an excellent fit.

Although RSM can statistically and mathematically suggest optimal solutions, these solutions are based on an individual characteristic, namely, a single response such as individual R_1 – R_7 , respectively. In case more than one observed response value of interest must be simultaneously optimized, and these observed response values of characteristics of interest exhibit a nonlinear function of design variables which must be presented in constraints, we then have to formulate the problem in a multiple objective manner and have them optimized concurrently. This can be done integrally via an optimization technique, such as mathematical programming (Arora, 2004). In this study,

multiple observed values of interest can be expressed by the R_1 – R_7 response functions listed in Table 20 of Appendix B. Any one of the response functions listed in Table 20 can be selected as the objective function of mathematical programming (MP). The remaining response functions with some forms of restraint are formulated as constraints in MP. Hence, there are seven possibilities of problem formulations which are dependent on the objectives R_1 – R_7 that we intend to optimize. The following MP formulation is an example with R_1 as an objective function for optimization. In order to formulate MOO by compromise programming, R_1 will be replaced with R_2 – R_7 , with same optimization constraints.

$$\text{Minimize } R_1 \quad (4)$$

$$\text{Subject to } R_1 \quad (5)$$

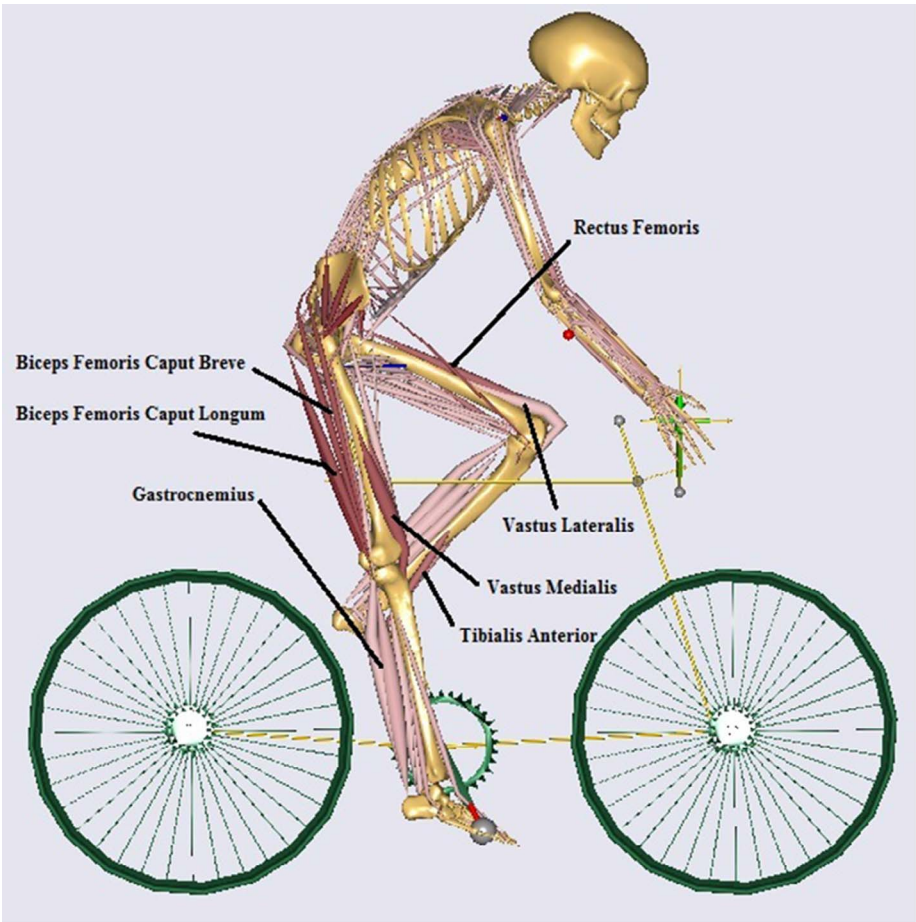


Fig. 5. The main muscles involved in proposed cycling analysis.

Table 1
The evaluation function linking the muscles involved during cycling.

Objective functions	Representative muscles
R ₁	Biceps femoris caput breve
R ₂	Biceps femoris caput Longum
R ₃	Gastrocnemius
R ₄	Rectus femoris
R ₅	Tibialis anterior
R ₆	Vastus lateralis
R ₇	Vastus medialis

Table 2
Three levels associated with product design variables.

Variables	Levels		
	Low	Medium	High
X ₁	75.280	77.280	79.280
X ₂	48.830	50.830	52.830
Y ₁	69.006	71.006	73.006
Y ₂	9.000	11.000	13.000

cm.

R₂ (6)

R₃ (7)

R₄ (8)

R₅ (9)

Table 3
The combined levels and the associated response values.

X ₁ X ₂ Y ₁ Y ₂	R ₁ R ₂ R ₃ R ₄ R ₅ R ₆ R ₇
Combine levels of design variables based on BBD experimental matrix for each trial	Response muscle activities from computer software AnyBody for various muscle groups under the corresponding trial

R₆ (10)

R₇ (11)

75.28 ≤ X₁ ≤ 79.28 (12)

48.83 ≤ X₂ ≤ 52.83 (13)

69.006 ≤ Y₁ ≤ 73.006 (14)

9 ≤ Y₂ ≤ 13 (15)

R₁ ≤ 50% (16)

R₂ ≤ 50% (17)

R₃ ≤ 50% (18)

R₄ ≤ 50% (19)

R₅ ≤ 50% (20)

R₆ ≤ 50% (21)

R₇ ≤ 50% (22)

(9) X₁,X₂,Y₁,Y₂ ≥ 0 and R_j ≥ 0 where j= 1...7

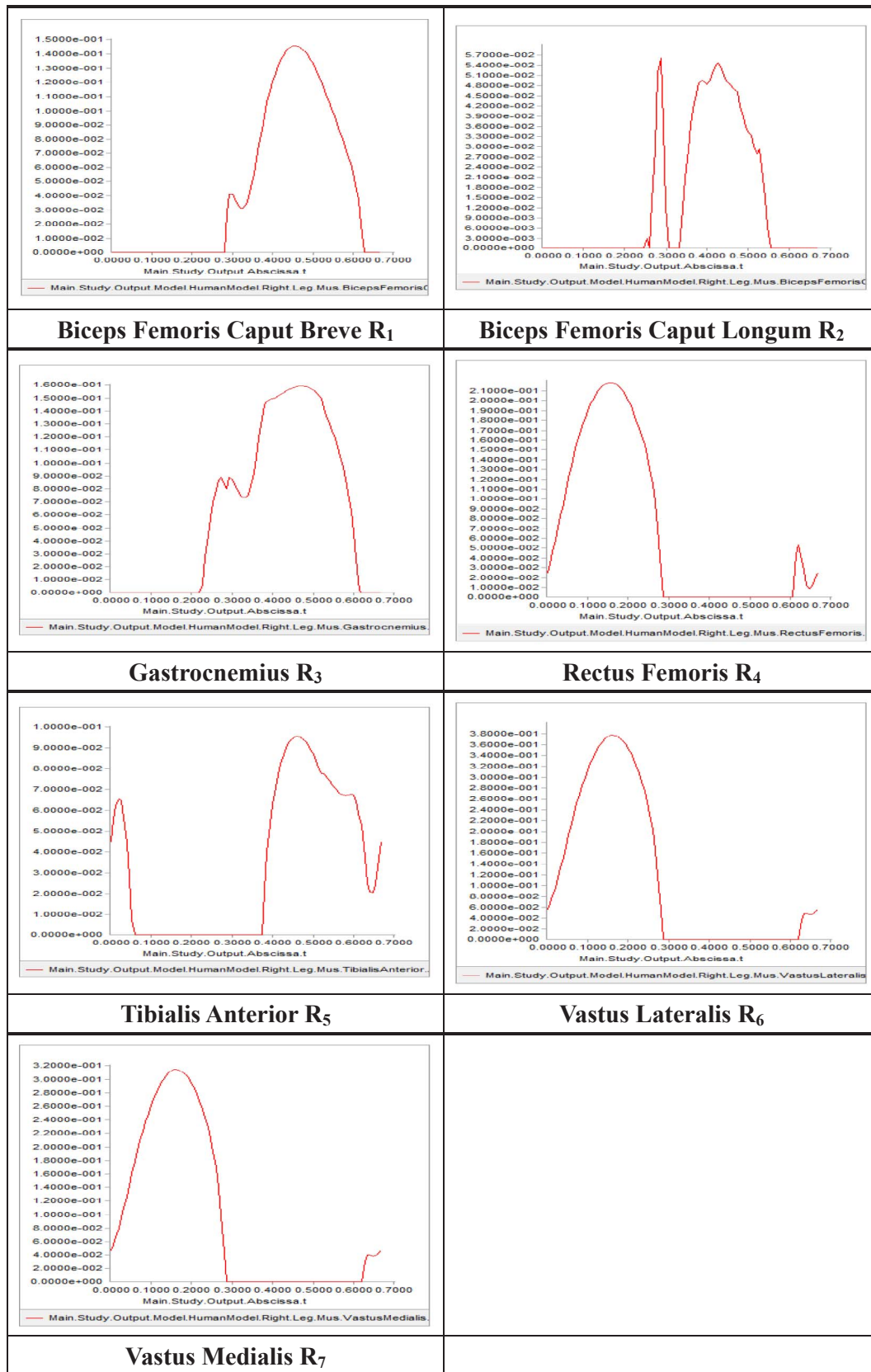


Fig. 6. The plot of muscle activities for R_1 – R_7 for the first run of Table 3.

Eqs. (5)–(11) are the response functions, R_1 – R_7 , which can be obtained from Table 20 of Appendix B. Although response function R_1 simultaneously appears in Eq. (4) as an objective function and Eq. (5) as the constraint, Eq. (5) does not have any influence on the feasible space

of the above MP formulation. That is, the optimal solutions are identical whether Eq. (5) is included or excluded in MP formulation. Obviously, the R_1 values of Eqs. (4) and (5) are identical at the optimal solutions of X_1 , X_2 , Y_1 , and Y_2 . The reason for setting the equations of R_1 – R_7 in

Table 4

The optimal solutions in minimization of single objective function R.

Variables and responses	Optimal values
X_1	75.2800 cm
X_2	48.8300 cm
Y_1	69.0060 cm
Y_2	9.0000 cm
R_1	6.90%
R_2	5.01%
R_3	10.49%
R_4	21.85%
R_5	5.23%
R_6	38.07%
R_7	31.40%

Note: % represents the muscle activity measurements.

constraints is to find the associated values simultaneously for convenience as MP optimization is completed. For example, when R_1 had the objective function, the found response values of R_1^* and R_2 – R_7 were given in Table 4 and rearranged in the first row of Table 5. With the same theme, when R_2 is considered as the objective function, the corresponding response values of R_1 , R_2^* , and R_3 – R_7 are listed in the second row of Table 5. The given lower and higher limits of X_1 , X_2 , Y_1 , and Y_2 are the values under the column of high and low levels in Table 2. Because the high muscle activity levels signify an increased risk of injury, there is a need to reduce the risk of injuries during product design. In this regard, Eqs. (16)–(22) establish the percentage upper levels of R_1 – R_7 , so that the cyclists' health and comfort under the decided design values are ensured. In this example, the upper bound of activity level is arbitrarily assigned as 50%. With the aid of GAMS software (Brooke, Kendrick, Meeraus, & Raman, 1998), the optimal solutions are listed in Table 4. As in the preceding illustration, there is a possibility that the remaining response functions R_2 , R_3 , R_4 , R_5 , R_6 , and R_7 , could also be of concern in a specific product design. The objective function of Eq. (4), R_1 , is then replaced with the other response functions under the same constraints as the above formulation. By following the notations defined in Eqs. (1)–(3), Table 5 is prepared for the multi-objective formulation by CP in the following subsection. We may see the R_1 row as an example for explaining Table 5. All the values shown in this row are correspondingly duplicated from the example of Table 4. Similar acts with different objective functions to obtain Table 4 as an example are then further used for forming the rest of the rows: R_2 , R_3 , R_4 , R_5 , R_6 , and R_7 in Table 5. The value, 6.64%* considered as R_1^* in CP formulation, shown in column R_1 of Table 5, is the least one (the most preferred) among R_1 , R_2 , R_3 , R_4 , R_5 , R_6 , and R_7 . Conversely, 39.23% is the largest one (the least preferred one) which is considered as

 $R_{1,\max}$

in CP. A similar approach is applied to the rest of the rows and columns to form the entire Table 5, which will be further used in the next subsection to formulate MOO via CP.

Table 5

The numerical results of a single objective for the nonrandom case.

Row	Column														
	(1) R ₁	r	(2) R ₂	r	(3) R ₃	r	(4) R ₄	r	(5) R ₅	r	(6) R ₆	r	(7) R ₇	r	Collective rank
min R ₁ (1)	6.90%	3	5.01%	3	10.49%	3	21.85%	6	5.23%	1	38.07%	6	31.40%	6	4
min R ₂ (2)	6.64%*	1	1.61%*	1	8.44%	2	21.40%	5	5.78%	3	37.62%	5	31.04%	5	2
min R ₃ (3)	6.83%	2	1.61%	2	8.42%*	1	21.35%	4	5.91%	4	37.51%	4	30.95%	4	1
min R ₄ (4)	38.58%	5	10.12%	5	50.00%	5	19.03%*	1	16.09%	5	32.63%	3	26.85%	3	3
min R ₅ (5)	6.90%	3	5.01%	3	10.49%	3	21.85%	6	5.23%*	1	38.07%	6	31.40%	6	4
min R ₆ (6)	39.23%	6	12.36%	6	50.00%	5	19.12%	2	16.84%	6	32.62%*	1	26.84%*	1	3
min R ₇ (7)	39.23%	6	12.36%	6	50.00%	5	19.12%	2	16.84%	6	32.62%*	1	26.84%*	1	3

Note: “r” represents ranking.

* Signify the associated value is the least value under corresponding column.

Table 6

The optimal solutions via compromised multi-objective formulation at upper level 50% (nonrandom case).

Variables	Optimal values	
X_1^*	79.2800 cm	
X_2^*	52.8300 cm	
Y_1^*	69.0060 cm	
Y_2^*	9.0000 cm	
Responses	Response values	Anybody values
R_1	13.63%	13.19%
R_2	7.03%	6.90%
R_3	27.83%	22.48%
R_4	19.27%	19.28%
R_5	4.41%	8.63%
R_6	33.65%	33.38%
R_7	27.71%	27.78%

Note: % represents the muscle activity.

6.1.1. Formulation for MOO via compromise programming

The method of compromise programming is adopted for formulating MOO with multiple muscle activity functions in the proposed approach.

$$T = \left| \frac{R_1 - R_1^*}{R_{1,\max} - R_1^*} \right| + \left| \frac{R_2 - R_2^*}{R_{2,\max} - R_2^*} \right| + \left| \frac{R_3 - R_3^*}{R_{3,\max} - R_3^*} \right| + \left| \frac{R_4 - R_4^*}{R_{4,\max} - R_4^*} \right| + \left| \frac{R_5 - R_5^*}{R_{5,\max} - R_5^*} \right| + \left| \frac{R_6 - R_6^*}{R_{6,\max} - R_6^*} \right| + \left| \frac{R_7 - R_7^*}{R_{7,\max} - R_7^*} \right| \quad (23)$$

In order to simplify the above formulation, the values associated with R_j^* and $R_{j,\max}$ are obtained from Table 15, where j is 1, 2, 3, ..., 7. They are:

$$R_1^* = 0.0664, \quad R_2^* = 0.0161, \quad R_3^* = 0.0842, \quad R_4^* = 0.1903, \quad R_5^* = 0.0523, \quad R_6^* = 0.3262, \quad R_7^* = 0.2684.$$

and

$$R_{1,\max} = 0.3923, \quad R_{2,\max} = 0.1236, \quad R_{3,\max} = 0.5000, \quad R_{4,\max} = 0.2185, \quad R_{5,\max} = 0.1684, \quad R_{6,\max} = 0.3807, \quad R_{7,\max} = 0.3140$$

The simplified expression of Eq. (23) is:

$$T = \frac{R_1 - 0.0664}{0.3259} + \frac{R_2 - 0.0161}{0.1075} + \frac{R_3 - 0.0842}{0.4158} + \frac{R_4 - 0.1903}{0.0282} + \frac{R_5 - 0.0523}{0.1162} + \frac{R_6 - 0.3262}{0.0544} + \frac{R_7 - 0.2684}{0.0456} \quad (24)$$

With Eq. (24) as the objective function and Eqs. (5)–(22) as the constraints, the optimal solutions and associated muscle activities were found and listed in the left-bottom of Table 6. The associated muscle activity plot for a cycle at the optimal solutions is also depicted in Fig. 7. To verify the validity of the proposed approach, the just found optimal values of designed variables X_1 , X_2 , Y_1 , and Y_2 were used to run

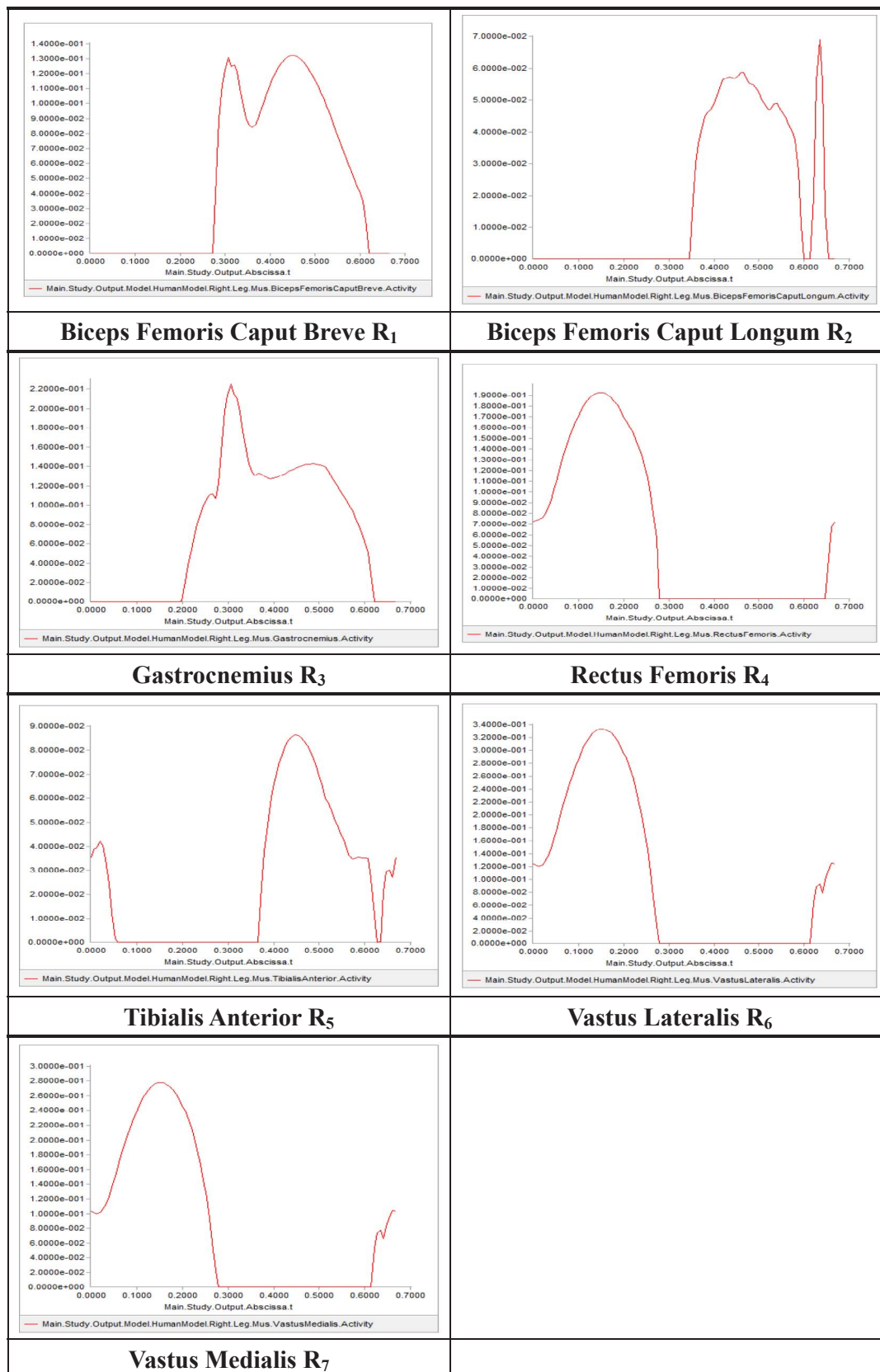


Fig. 7. The plot of muscle activities for R_1 – R_7 at the optimal design values.

AB to obtain the muscle activities located at the right-bottom column of Table 6. The comparison between the muscle activities shown in the left-lower column and right-lower column of Table 6 shows no significant difference. Thus, the validity of the proposed approach is

acceptable. Although the limit of muscle activity in Eqs. (16)–(22) is set at only a 50% upper level, there are other possible limits which represent various levels of health and comfort required for satisfying different cyclists' expectations. Thus, a sensitivity analysis performed

Table 7

The numerical results of a single objective (random case).

Row	Column														
	R ₁ (1)	r	R ₂ (2)	r	R ₃ (3)	r	R ₄ (4)	r	R ₅ (5)	r	R ₆ (6)	r	R ₇ (7)	r	Collective rank
min R ₁ (1)	11.99%	2	5.65%	2	10.46%	2	20.86%	6	09.57%	6	37.39%	6	31.27%	6	5
min R ₂ (2)	11.53%	1	2.53%	1	10.76%	3	19.69%	5	09.24%	5	35.23%	5	29.47%	5	3
min R ₃ (3)	12.72%	3	6.06%	3	9.43%	1	21.46%	7	10.07%	7	39.04%	7	32.64%	7	6
min R ₄ (4)	14.68%	5	7.09%	4	17.93%	5	18.52%	1	07.32%	2	31.78%	3	26.63%	3	1
min R ₅ (5)	13.69%	4	7.22%	5	15.44%	4	18.57%	2	07.20%	1	31.88%	4	26.71%	4	2
min R ₆ (6)	16.69%	6	7.26%	6	22.94%	6	18.58%	3	07.60%	3	31.69%	1	26.55%	1	4
min R ₇ (7)	16.69%	6	7.26%	6	22.94%	6	18.58%	3	07.60%	3	31.69%	1	26.55%	1	4

Note: “r” represents ranking.

* Signify the least value under corresponding column.

Table 8

The optimal solutions via compromised multi-objective formulation at the upper level 50% (random case).

Controllable variables	Optimal values	
X ₁ [*]	78.1864 cm	
X ₂ [*]	50.6260 cm	
Y ₁ [*]	69.0060 cm	
Y ₂ [*]	9.0000 cm	
Responses	Response at optimal values	AnyBody at optimal values
R ₁	13.64%	13.35%
R ₂	5.77%	5.99%
R ₃	15.23%	14.86%
R ₄	19.70%	19.92%
R ₅	8.87%	8.59%
R ₆	34.34%	34.23%
R ₇	28.75%	28.48%

Note: % represents the muscle activity.

for the nonrandom case under various upper levels: 30%, 35%, 40%, 45%, and 50% was completed and is listed in Table 9. The magnitudes of the percentages of the upper levels represent the levels of cyclists' tolerance in accepting the risk of possible injuries. For example, the lower upper level, 30%, in the constraints indicates that the cyclists are less tolerant about facing possible injuries. That is, the cyclists' attitude about injury and safety for a product design is one of risk aversion. Conversely, for example, a higher percentage upper level, such as 50%, in the constraints represents the rider's mind-set as a prone risk attitude. For the other example, the cyclists may ride in extreme or unknown conditions. The percentages of the upper levels have to be established at higher levels so that possible injuries can be reduced in

advance by robust product design for uncertain or severe environments. The percentages of the upper levels do not necessarily have to be identical values; they can vary for additional considerations.

6.2. For randomized height, weight of cyclists case

By following the ANOVA and RSM regression analysis as done in the nonrandom case of Section 6.1, we obtained Tables 21 and 22 of Appendix C, as well as Tables 7, 8 and Fig. 8 for the random case. Table 21 summarizes the individual F-ratios, importance rankings, and combined importance rankings. This information provides the direction of design improvement. For example, the corresponding score of X₁ in last column is 9, which is the least one among the four controllable variables. The least score indicates that the combined ranking of X₁ is the most important controllable variable to which we have to pay attention during the product design improvement. As shown in the last column of Table 21, the order of combined importance for design variables is X₁, X₂, Y₁, and Y₂. Obviously, as compared to the nonrandom case in Table 19, X₁ is still the most important design variable for the random case. By looking into the scores in last column of Table 19 and Table 21, the deviation between X₂'s and Y₁'s scores of Table 21 is smaller than the deviation between X₂'s and Y₁'s scores in Table 19. This consequence indicates that there is a possibility that design variable Y₁ under the random case will become more important than X₂ under the nonrandom case for the coming product design improvement. Table 22 of Appendix C presents the response functions for the next MOO formulation. Table 7 in the random case provides the single objective value, like Table 5 preparing for MOO formulation via CP in the non-random case. The optimal solutions of MOO are listed in Table 8. Similar to the nonrandom case, the sensitivity analyses examining various risk levels of injury occurrence for a product design are also performed

Table 9

The optimal solutions at upper levels 30%, 35%, 40%, 45%, and 50%.

Controllable variables, responses, and T value	Upper level 30% Optimal values		Upper level 35% Optimal values		Upper level 40% Optimal values		Upper level 45% Optimal values		Upper level 50% Optimal values	
	Non-random	Random	Non-random	Random	Non-random	Random	Non-random	Random	Non-random	Random
X ₁ ^{**}	78.1033 cm	79.2800 cm	79.2800 cm	79.2800 cm	79.0560 cm	78.1864 cm	79.1599 cm	78.1864 cm	79.2800 cm	78.1864 cm
X ₂ ^{**}	48.8300 cm	48.8300 cm	52.8300 cm	48.8300 cm	52.8300 cm	50.6260 cm	52.8300 cm	50.6260 cm	52.8300 cm	50.6260 cm
Y ₁ ^{**}	70.7504 cm	69.8933 cm	69.0060 cm	69.0060 cm	69.0060 cm	69.0060 cm	69.0060 cm	69.0060 cm	69.0060 cm	69.0060 cm
Y ₂ ^{**}	11.0650 cm	11.7958 cm	9.0000 cm	13.0000 cm	9.0000 cm	9.0000 cm	9.0000 cm	9.0000 cm	9.0000 cm	9.0000 cm
R ₁ ^{**}	30.00%	16.69%	13.63%	13.69%	12.48%	13.64%	12.99%	13.64%	13.63%	13.64%
R ₂ ^{**}	10.55%	7.26%	7.03%	7.22%	6.50%	5.77%	6.73%	5.77%	7.03%	5.77%
R ₃ ^{**}	30.00%	22.94%	27.83%	15.44%	24.87%	15.23%	26.20%	15.23%	27.83%	15.23%
R ₄ ^{**}	19.96%	18.58%	19.27%	18.57%	19.42%	19.70%	19.35%	19.70%	19.27%	19.70%
R ₅ ^{**}	15.62%	7.60%	4.41%	7.20%	4.51%	8.87%	4.46%	8.87%	4.41%	8.87%
R ₆ ^{**}	30.00%	30.00%	33.65%	31.88%	33.95%	34.34%	33.81%	34.34%	33.65%	34.34%
R ₇ ^{**}	28.05%	26.55%	27.71%	26.71%	27.96%	28.75%	27.85%	28.75%	27.71%	28.75%
T ^{**}	5.7386	1.5971	2.1678	1.8702	1.7090	3.2272	1.6403	3.2272	1.5802	3.2272

** Represents the optimal values of multi-objective optimization via compromise programming.

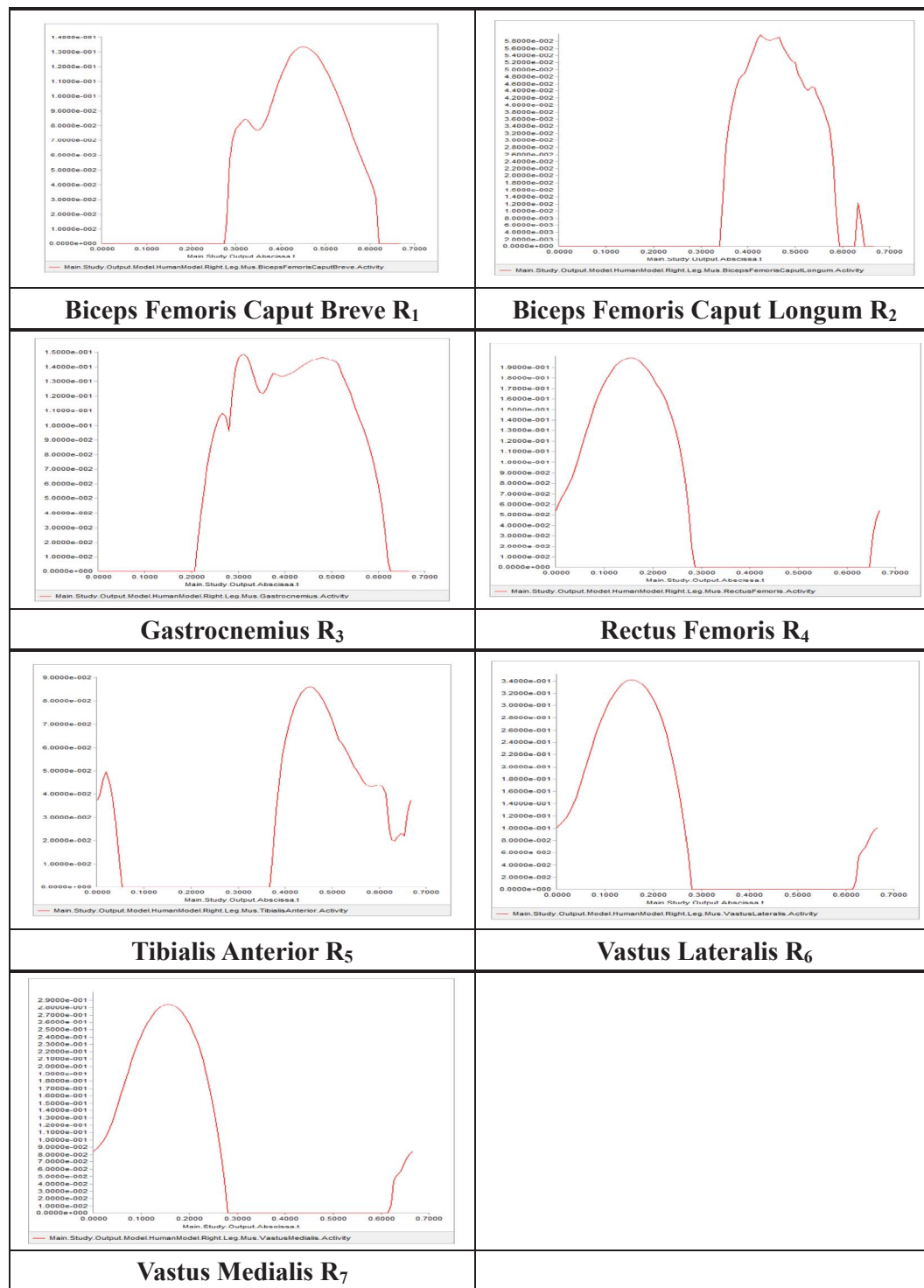


Fig. 8. The plot of muscle activities for R₁-R₇ at the optimal design values for random case at upper level 50%.

under a range of upper levels 30%, 35%, 40%, 45%, and 50% for the random case. The results of sensitivity analyses for the nonrandom and random cases in the range of 30%-50% are expressed simultaneously in Table 9.

The T values shown in Table 9 constructed from the CP formulation for minimization represent the collective evaluation scores under various sets of designed parameter values. Namely, an analysis of T values can help us to make right decisions in determining optimal product parameter values. The T values for the nonrandom case are greater than for the random case under the 35% upper level. However, the T values for the random case become greater than the value for the nonrandom

case above the 35% upper level. This phenomenon suggests that the product design with the random case is superior to the nonrandom case under the 35% upper level. Conversely, the product design with the random case is inferior to the nonrandom case above the 35% upper level. As is known, the magnitude of the percentage of the upper level represents the cyclist's tolerable risk level when the cyclists ride the designed bicycles. Thus, if the cyclist's attitude is risk prone, a high percentage of the upper level, the product design with the nonrandom case is a good choice. Conversely, the product design with the random case is an appropriate selection when the cyclist's attitude is one of risk aversion, a low percentage of the upper level. Theoretical speaking the

product design with random case is more near the true design during realistic application than with nonrandom case. Thus, even the product design with non-random case is a good choice in this particular example, the product design with random case should be considered as a good reference under uncertain % upper level and working conditions.

In order to make the results of Table 9 more comprehensible, they are plotted in Fig. 9 for further explanation which can be considered as references during product design. The observations are listed in the following:

1. Generally, for 30–50% the optimal parameter values of X_2 , Y_1 , R_2 , and R_3 for the nonrandom case are greater than the corresponding values for the random case. Because a higher muscle activity level usually results in a higher risk of MSD injury for a cyclist, it is wise to design a product with a random case. To compensate for the high risk under the product design with a nonrandom case, the high optimal product design values of X_2 and Y_1 resulted. Conversely, the determined value of Y_2 for a product design with non-randomness is less than the values determined with randomness.
2. The values of R_4 , R_5 , R_6 , and R_7 are greater for the nonrandom case than for the random case under the upper level in the range of 35–40%. However, above the range of 35–40%, these values are

smaller for the nonrandom case than for the random case. The value of X_1 for the nonrandom case is less compared to the random case under the 35% upper level. Conversely, above the 35% upper level, the value of X_1 for the nonrandom case is greater than the values for the random case.

3. At the 30% upper level, the R_1 muscle activity measurement is significantly greater for the nonrandom case than for the random case. The R_1 muscle activity measurement is slightly less for the nonrandom case than for the random case in the range of 35–50%. The muscle activity measurement involved for R_4 , R_5 , R_6 , and R_7 is greater for the nonrandom case than for the random case under the ranges of 35–40% upper level. Conversely, the muscle activity measurements involved for R_4 , R_5 , R_6 , and R_7 become less for the nonrandom case than for the random case above the upper level ranges of 35–40% upper level.
4. Another important feature about inclinations concluded from Fig. 9 is that the magnitudes of X_1 , X_2 , Y_1 , Y_2 , R_1 – R_7 and T are always altered in the upper level 35–40%. The clear alternation at this upper level represents the associated optimal determined values are very sensitive to the cyclists' attitude when shifting from risk aversion (below upper level 35–40%) to risk prone (above upper level 35–40%).

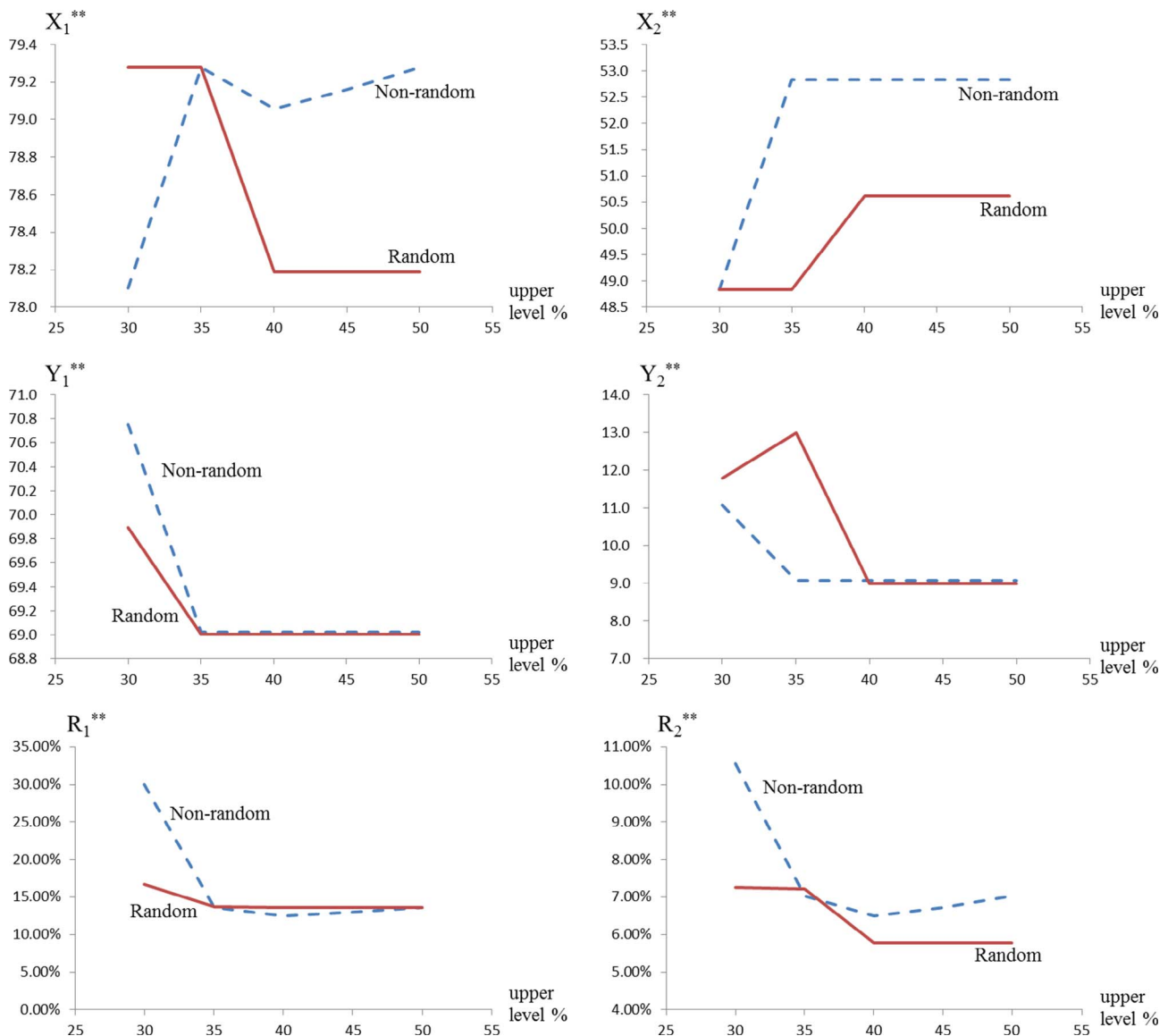


Fig. 9. The plot of optimal parameter values of X_1 , X_2 , Y_1 , and Y_2 and muscle activity levels, R_1 – R_7 at upper levels, 30–50% for random case and nonrandom case.

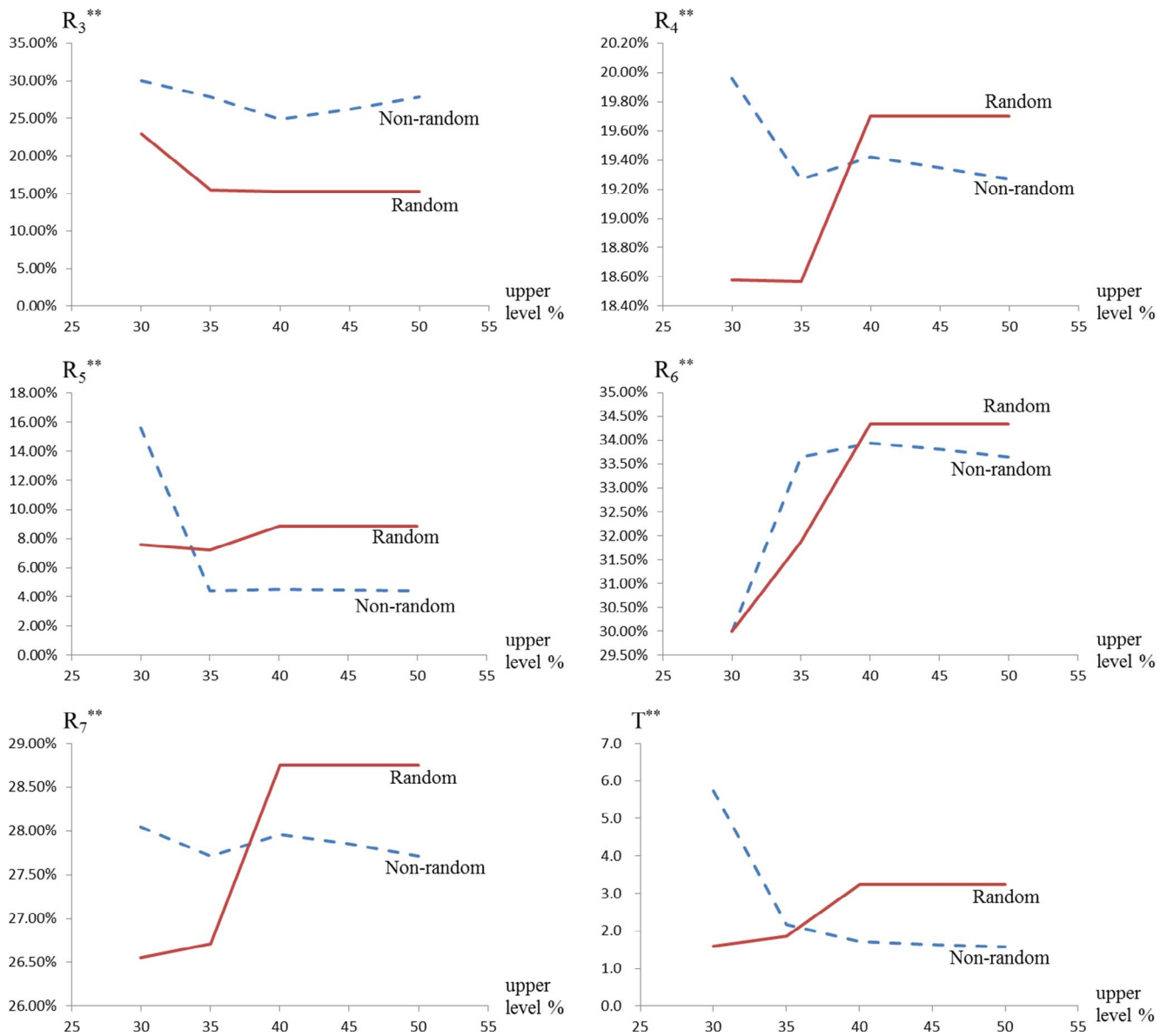


Fig. 9. (continued)

As aforementioned, the percentage of the upper level is established to represent how much risk of injury the cyclists are willing to face when riding a bike. Namely, the higher percentages of the upper levels symbolize a higher acceptance of possible injury for the cyclists under given design parameter values. In addition, it is known that the risk of MSD injury is proportional to the muscle activity measurements. Thus, when intense conditions do not exist, it is a wise decision to select the product parameter values with low percentages of the upper levels and low muscle activity measurements, R_1 – R_7 , for conservative consideration. That is, the optimal product parameter values are the ones under the random case and 35% upper level.

Table 10 summarizes the rates of change for the depicted rising or falling inclination in Fig. 9. The rates of change in Table 10 can be used as references for the decision-making process during the product design. The terms: *constant*, *mild* and *sharp* describe the levels of the rates of change for the design parameter values and muscle activity measurements with respect to the percentages of the upper levels. The rate of change also indicates the level of sensitivity regarding the decided product parameter value under varied percentages of the upper levels. Thus, the degree of the rate of change can help the designers establish the risk level such as the percentages of

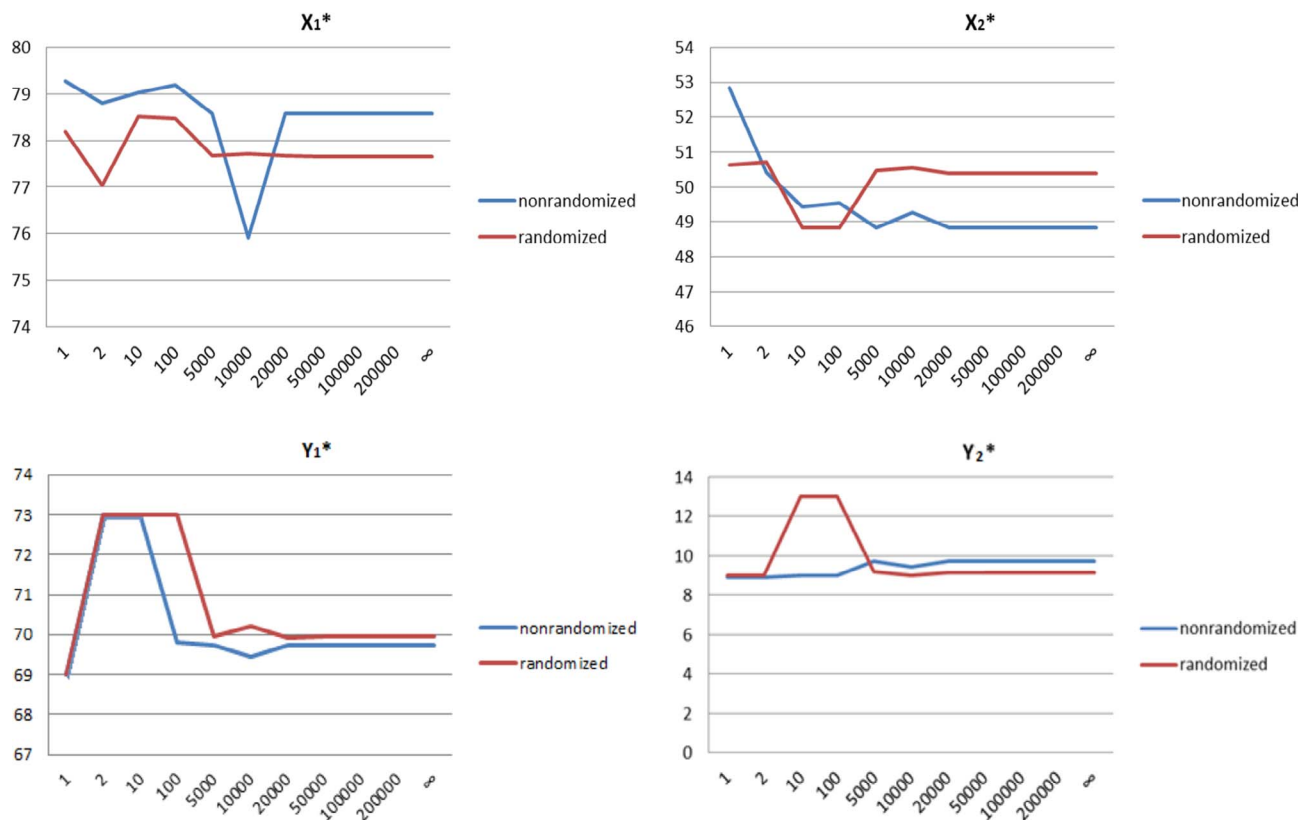
the upper levels for users' tolerance regarding possible injuries. Thus, for robustness consideration, our preference is for the constant or mild inclination of optimal response values and product parameter values under the selected case and the percentages of the upper level. For the presented example, the product parameter values with random case and 30–35% upper level are the most robust decisions among all the possible selections.

When product design is based on the optimal product parameter values determined at high percentages of the upper level, the cyclists' injuries can be reduced either under the expected or unexpected extreme riding conditions. Although a product design established at high percentages of the upper level can adapt to the extreme working conditions, the resulting muscle activity measurements, R_1 – R_7 , are usually high enough for injuries to occur sometimes. Conversely, for a product design found at the low percentages of the upper level for normal working conditions, the resulting muscle activity measurements, R_1 – R_7 , are lowly sufficient to reduce the possible injuries. However, the designed values found from the low percentages of the upper level are appropriate only when the environment is normal. As aforementioned, the risks can be magnified when cycling under severe road conditions, bad weather conditions, and long periods of time. In order to

Table 10

The trend of the plots depicted in Fig. 9.

Controllable variables, responses, and T values		30–35%	35–40%	40–45%	45–50%
X_1^{**}	Non-random	Sharply increasing	Mildly decreasing	Mildly increasing	Mildly increasing
	Random	Constant	Sharply decreasing	Constant	Constant
X_2^{**}	Non-random	Sharply increasing	Constant	Constant	Constant
	Random	Constant	Sharply increasing	Constant	Constant
Y_1^{**}	Non-random	Sharply decreasing	Constant	Constant	Constant
	Random	Mildly decreasing	Constant	Constant	Constant
Y_2^{**}	Non-random	Sharply decreasing	Constant	Constant	Constant
	Random	Mildly increasing	Sharply decreasing	Constant	Constant
R_1^{**}	Non-random	Sharply decreasing	Very mildly decreasing	Very mildly increasing	Very mildly increasing
	Random	Mildly decreasing	Constant	Constant	Constant
R_2^{**}	Non-random	Sharply decreasing	Mildly decreasing	Mildly increasing	Mildly increasing
	Random	Constant	Sharply decreasing	Constant	Constant
R_3^{**}	Non-random	Very mildly decreasing	Very mildly decreasing	Very mildly increasing	Very mildly increasing
	Random	Mildly decreasing	Constant	Constant	Constant
R_4^{**}	Non-random	Mildly decreasing	Very mildly increasing	Very mildly decreasing	Very mildly decreasing
	Random	Constant	Sharply increasing	Constant	Constant
R_5^{**}	Non-random	Sharply decreasing	Constant	Constant	Constant
	Random	Very mildly decreasing	Mildly increasing	Constant	Constant
R_6^{**}	Non-random	Sharply increasing	Mildly increasing	Very mildly decreasing	Very mildly decreasing
	Random	Sharply increasing	Sharply increasing	Constant	Constant
R_7^{**}	Non-random	Mildly decreasing	Mildly increasing	Very mildly decreasing	Very mildly decreasing
	Random	Mildly increasing	Sharply increasing	Constant	Constant
T^{**}	Non-random	Sharply decreasing	Very mildly decreasing	Very mildly decreasing	Very mildly decreasing
	Random	Very mildly increasing	Mildly increasing	Constant	Constant

**Fig. 10.** The plot of optimal parameter values of X_1 , X_2 , Y_1 and Y_2 and muscle activity levels, R_1 – R_7 , at upper levels 50% and different q values for the random case and nonrandom case.

compensate for the users' application in extreme or uncontrollable environments, the percentages of the upper levels in Eqs. (16)–(22) can be adjusted appropriately at high values for conservative product design. Thus, the sensitivity analysis for particular percentages of the

upper level, as done in Table 9 and Fig. 9, can help the designers establish the percentages of the upper level for users' willingness to accept the risk of possible injuries under known or unknown working conditions.

It is known that the q values of CP range between 1 and ∞ . The large magnitude of q values represents the great penalty resulting from the deviations between the found optimal solution and ideal optimal solution. That is, a large value of q will force the found optimal solution closer to the ideal optimal solution. In this regard, sensitivity analysis for various values of q is performed for MOO in a range between 1 and ∞ . The analyzed results are given in Tables 23 and 24 of Appendix D. These results are also plotted in Fig. 10 to aid the reader's further comprehension. The optimal values of decision variables, X_1 , X_2 , Y_1 , and Y_2 rise and fall along the q value from 1 to ∞ and eventually converge to a value as q approaches a great value. It is clear that the optimal values when q is 1 are very close to the optimal values when q is very great, particularly for random cases in the presented example. As in the aforementioned discussion, the proposed approach is mainly and mostly applicable for product design under uncertainty. However, the above conclusion about q values may not always be true for other examples. An analysis for deriving optimal solutions for q values under a range from 1 to ∞ has to be run in advance before the next analysis.

7. Discussion

The muscles are activated by the Central Nervous System (CNS) which coordinates the muscles to achieve precise and harmonious movements. There are more muscles than necessary to impel the degrees of freedom of the musculoskeletal system. From the dynamic point of view, there are many possibilities of muscle coordination for adequate exercises. The problem of muscle coordination is often considered as redundancy modeling of muscle employment. The modeling of these mechanisms is based on the assumption that the activated muscles for appropriate coordination are achieved by the Central Nervous System (CNS), typically with inherent optimality. The technique of MOO considered in the proposed approach is based on the above optimality assumption.

The World Health Organization proposed that disorders of the musculoskeletal system characterize the major source of employee absence from work. The disorders result in employers' additional cost because of unexpected employee absence, the individuals' loss of

income during such absence, and the personal expenditure on medication during the course of injury treatment (Luttman et al., 2003; Niu, 2010). The risk factors of the musculoskeletal-related disorders include constant or uncomfortable postures, usage pace or use/rest cycle, individual health, physical conditions, age, etc. In addition to a variety of risk factors involved, the uncertainties regarding associated risk factor magnitudes have to be considered as well, in order to understand the problem of WMSDs (Oakman & Chan, 2015). In other words, the conventional approach to establishing guidelines through a series of interviews to reduce WMSDs is difficult to realize (Anton & Weeks, 2016; AyuBidiawati & Suryani, 2015; Azadeh et al., 2015; Chung & Choi, 1997; Sanderson & Amoroso, 2009). In contrast to the conventional approaches, an ergonomic promotional work design for a multi-dimensional approach with uncertainties to prevent users from work-related musculoskeletal disorders (WMSDs) is needed (Oakman & Chan, 2015). The proposed approach for an ergonomic product design can also be used for job designs to realize WMSDs-free working environments.

8. Conclusion

Through the incorporation of AB computer software, the RSM statistical method and compromised MOO technique, the proposed approach for product design was developed herein. With the presented approach, the optimal product parameters can be found for the best ergonomic fit between users and products. Additionally, with the statistical method in the presented approach, the importance rankings can be identified as reference for product design improvement. Because of statistical optimization and randomized users' sizes with upper level muscle activities constrained, the discovered product parameter values are robust to the variation of users' size under the tolerable risk of possible injuries. Owing to the computer musculoskeletal model, the product design can also be completed in the least time and cost at the earliest stage of development. In addition, users can select their most suitable fit of product via computer musculoskeletal modeling in real time instead of the conventional questionnaire for suitable product selection.

Appendix A. Experimental design matrix of RSM

Table 11

Table 11
Experimental design matrix of Box-Behnken design.

No.	X_1	X_2	Y_1	Y_2
1	-1	-1	0	0
2	1	-1	0	0
3	-1	1	0	0
4	1	1	0	0
5	0	0	-1	-1
6	0	0	1	-1
7	0	0	-1	1
8	0	0	1	1
9	-1	0	0	-1
10	1	0	0	-1
11	-1	0	0	1
12	1	0	0	1
13	0	-1	-1	0
14	0	1	-1	0
15	0	-1	1	0
16	0	1	1	0
17	-1	0	-1	0
18	1	0	-1	0
19	-1	0	1	0
20	1	0	1	0

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Table 11 (continued)

No.	X ₁	X ₂	Y ₁	Y ₂
21	0	−1	0	−1
22	0	1	0	−1
23	0	−1	0	1
24	0	1	0	1
25	0	0	0	0
26	0	0	0	0
27	0	0	0	0

Factors = 4, runs = 27.

Appendix B. Statistical analyses for nonrandom case via RSM

Tables 12–20

Table 12

Statistical analysis for muscle activity R₁.

Response surface for variable R ₁					
Response mean		0.177128			
Root MSE		0.068300			
R-Square		0.7814			
Coef. of variation		38.5597			
Regression	D.F.	S.S.	R-Square	F-Ratio	Prob > F
Linear	4	0.070221	0.329	3.76	0.0406
Quadratic	4	0.04335	0.2031	2.32	0.1275
Cross product	6	0.053199	0.2493	1.9	0.1763
Total regress	14	0.166769	0.7814	2.55	0.0705
Residual	D.F.	S.S.	M.S.	F-Ratio	Prob > F
Lack of fit	10	0.046649	0.004665		
Pure error	0	0			
Total error	10	0.046649	0.004665		
Factor	D.F.	S.S.	M.S.	F-Ratio	Prob > F
X ₁	5	0.127245	0.025449	5.46	0.0112
X ₂	5	0.068032	0.013606	2.92	0.0704
Y ₁	5	0.000308	0.0000616	0.01	0.9999
Y ₂	5	0.000273	0.0000546	0.01	0.9999

Table 13

Statistical analysis for muscle activity R₂.

Response surface for variable R ₂					
Response mean		0.063944			
Root MSE		0.013352			
R-Square		0.8963			
Coef. of variation		20.8815			
Regression	D.F.	S.S.	R-Square	F-Ratio	Prob > F
Linear	4	0.008839	0.514	12.39	0.0007
Quadratic	4	0.004608	0.268	6.46	0.0078
Cross product	6	0.001966	0.1144	1.84	0.1884
Total regress	14	0.015413	0.8963	6.18	0.0032
Residual	D.F.	S.S.	M.S.	F-Ratio	Prob > F
Lack of fit	10	0.001783	0.000178		
Pure error	0	0			
Total error	10	0.001783	0.000178		
Factor	D.F.	S.S.	M.S.	F-Ratio	Prob > F
X ₁	5	0.00962	0.001924	10.79	0.0009
X ₂	5	0.003072	0.000614	3.45	0.0454
Y ₁	5	0.001846	0.000369	2.07	0.1531
Y ₂	5	0.000238	0.0000476	0.27	0.921

Table 14
Statistical analysis for muscle activity R₃.

Response surface for variable R ₃					
Response mean		0.212420			
Root MSE		0.050881			
R-Square		0.9255			
Coef. of variation		23.9531			
Regression	D.F.	S.S.	R-Square	F-Ratio	Prob > F
Linear	4	0.176844	0.5088	17.08	0.0002
Quadratic	4	0.106492	0.3064	10.28	0.0014
Cross product	6	0.038365	0.1104	2.47	0.0991
Total regress	14	0.321701	0.9255	8.88	0.0007
Residual	D.F.	S.S.	M.S.	F-Ratio	Prob > F
Lack of fit	10	0.025889	0.002589		
Pure error	0				
Total error	10	0.025889	0.002589		
Factor	D.F.	S.S.	M.S.	F-Ratio	Prob > F
X ₁	5	0.252907	0.050581	19.54	< .0001
X ₂	5	0.048359	0.009672	3.74	0.0362
Y ₁	5	0.000291	0.0000582	0.02	0.9997
Y ₂	5	0.000141	0.0000281	0.01	1

Table 15
Statistical analysis for muscle activity R₄.

Response surface for variable R ₄					
Response mean		0.205604			
Root MSE		0.000162			
R-Square		0.9999			
Coef. of variation		0.0787			
Regression	D.F.	S.S.	R-Square	F-Ratio	Prob > F
Linear	4	0.002119	0.999	20246.9	< .0001
Quadratic	4	0.000000438	0	0.42	0.7921
Cross product	6	0.000001907	0.0009	12.15	0.0004
Total regress	14	0.002121	0.9999	5790.16	< .0001
Residual	D.F.	S.S.	M.S.	F-Ratio	Prob > F
Lack of fit	10	0.000000262	2.6166667E-8		
Pure error	0				
Total error	10	0.000000262	2.6166667E-8		
Factor	D.F.	S.S.	M.S.	F-Ratio	Prob > F
X ₁	5	0.0020850	0.000417	15934.6	< .0001
X ₂	5	0.0000050	0.000000990	37.84	< .0001
Y ₁	5	0.0000145	0.000002901	110.86	< .0001
Y ₂	5	0.0000188	0.000003755	143.49	< .0001

Table 16
Statistical analysis for muscle activity R₅.

Response surface for variable R ₅					
Response mean		0.101004			
Root MSE		0.048508			
R-Square		0.5256			
Coef. of variation		48.0259			
Regression	D.F.	S.S.	R-Square	F-Ratio	Prob > F
Linear	4	0.004534	0.0914	0.48	0.749
Quadratic	4	0.007315	0.1475	0.78	0.5646
Cross product	6	0.014221	0.2867	1.01	0.4715
Total regress	14	0.026071	0.5256	0.79	0.665
Residual	D.F.	S.S.	M.S.	F-Ratio	Prob > F

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Table 16 (continued)

Response surface for variable R ₅					
Lack of fit	10	0.023530	0.002353		
Pure error	0				
Total error	10	0.023530	0.002353		
Factor	D.F.	S.S.	M.S.	F-Ratio	Prob > F
X ₁	5	0.016999	0.0034	1.44	0.2896
X ₂	5	0.018249	0.00365	1.55	0.2591
Y ₁	5	0.0003	0.000060017	0.03	0.9996
Y ₂	5	0.000273	0.000054693	0.02	0.9997

Table 17Statistical analysis for muscle activity R₆.

Response surface for variable R ₆					
Response mean		0.355664			
Root MSE		0.000153			
R-Square		1.0000			
Coef. of variation		0.0430			
Regression	D.F.	S.S.	R-Square	F-Ratio	Prob > F
Linear	4	0.00964	0.9995	102,913	< .0001
Quadratic	4	0.000003924	0.0004	41.9	< .0001
Cross product	6	0.000000998	0.0001	7.1	0.0037
Total regress	14	0.009644	1	29418.7	< .0001
Residual	D.F.	S.S.	M.S.	F-Ratio	Prob > F
Lack of fit	10	0.000000234	2.3416667E−8		
Pure error	0				
Total error	10	0.000000234	2.3416667E−8		
Factor	D.F.	S.S.	M.S.	F-Ratio	Prob > F
X ₁	5	0.009215	0.001843	78706.3	< .0001
X ₂	5	0.000422	0.000084372	3603.06	< .0001
Y ₁	5	0.000003261	0.000000652	27.85	< .0001
Y ₂	5	0.000002942	0.000000588	25.13	< .0001

Table 18Statistical analysis for muscle activity R₇.

Response surface for variable R ₇					
Response mean		0.295976			
Root MSE		0.000146			
R-Square		1.0000			
Coef. of variation		0.0494			
Regression	D.F.	S.S.	R-Square	F-Ratio	Prob > F
Linear	4	0.006681	0.9995	77994.1	< .0001
Quadratic	4	0.000002651	0.0004	30.94	< .0001
Cross product	6	0.000000623	0.0001	4.84	0.0143
Total regress	14	0.006685	1	22,295	< .0001
Residual	D.F.	S.S.	M.S.	F-Ratio	Prob > F
Lack of fit	10	0.000000214	2.1416667E−8		
Pure error	0				
Total error	10	0.000000214	2.1416667E−8		
Factor	D.F.	S.S.	M.S.	F-Ratio	Prob > F
X ₁	5	0.006386	0.001277	59638.1	< .0001
X ₂	5	0.000293	0.000058686	2740.22	< .0001
Y ₁	5	0.000002249	0.00000045	21.01	< .0001
Y ₂	5	0.000001955	0.000000391	18.26	< .0001

Table 19
F-ratio and important ranking for $R_1 \sim R_7$ (nonrandom case).

Factors	R ₁ (1)		R ₂ (2)		R ₃ (3)		R ₄ (4)		R ₅ (5)		R ₆ (6)		R ₇ (7)		Combined important ranking (1) + (2) + (3) + (4) + (5) + (6) + (7)
	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	
X ₁	5.46	1	10.79	1	19.54	1	15934.6	1	1.44	2	78706.3	1	59638.1	1	8
X ₂	2.92	2	3.45	2	3.74	2	37.84	4	1.55	1	3603.06	2	2740.22	2	15
Y ₁	0.01	3	2.07	3	0.02	3	110.86	3	0.03	3	27.85	3	21.01	3	21
Y ₂	0.01	3	0.27	4	0.01	4	143.49	2	0.02	4	25.13	4	18.26	4	25

Table 20
Response functions of $R_1 \sim R_7$ (nonrandom case).

Factor	R-Square	Response functions
R_1	0.7814	$8.688149 - 1.640523X_1 + 1.75708X_2 + 0.243468Y_1 + 0.11596Y_2 + 0.019907X_1^2 - 0.0288X_1X_2 + 0.004442X_2^2 + 0.001006X_1Y_1 - 0.002258Y_1^2 - 0.000881X_1Y_2 - 0.00000625X_2Y_2 - 0.00218Y_2^2$
R_2	0.8963	$20.925289 - 0.799182X_1 + 0.380557X_2 + 0.006441Y_1 + 0.002658Y_2 + 0.006609X_1^2 - 0.005431X_1X_2 + 0.000488X_2^2 + 0.000956X_1Y_1 - 0.000219X_2Y_1 - 0.000563Y_1^2 - 0.0003X_1Y_2 + 0.00001875X_2Y_2 + 0.000419Y_1Y_2 - 0.000366Y_2^2$
R_3	0.9244	$104.560292 - 3.868366X_1 + 1.780238X_2 - 0.075819Y_1 + 0.115777Y_2 + 0.032619X_1^2 - 0.024362X_1X_2 + 0.000894X_2^2 + 0.001969X_1Y_1 - 0.00004375X_2Y_1 - 0.000531Y_1^2 - 0.001431X_1Y_2 + 0.00000625X_2Y_2 + 0.00005Y_1Y_2 - 0.000419Y_2^2$
R_4	0.9999	$1.838728 - 0.021415X_1 - 0.005195X_2 - 0.012723Y_1 + 0.007229Y_2 + 0.000014583X_1^2 + 0.00005X_1X_2 + 0.000002083X_2^2 + 0.00015X_1Y_1 + 0.000025X_2Y_1 - 0.000004167Y_1^2 - 0.00005625X_1Y_2 - 0.00003125X_2Y_2 - 0.00000625Y_1Y_2 - 0.000010417Y_2^2$
R_5	0.5256	$-28.001991 - 0.024559X_1 + 0.636502X_2 + 0.352485Y_1 + 0.054783Y_2 + 0.005111X_1^2 - 0.014906X_1X_2 + 0.004974X_2^2 - 0.00003125X_1Y_1 + 0.00003125X_2Y_1 - 0.00247Y_1^2 + 0.00001875X_1Y_2 - 0.00001875X_2Y_2 - 0.00001875Y_1Y_2 - 0.002457Y_2^2$
R_6	1.0000	$3.246672 - 0.050557X_1 - 0.000681X_2 - 0.013512Y_1 + 0.004704Y_2 + 0.00019X_1^2 + 0.0000375X_1X_2 - 0.000041667X_2^2 + 0.00008125X_1Y_1 + 0.000075X_2Y_1 + 0.000023958Y_1^2 - 0.000025X_1Y_2 - 0.00003125X_2Y_2 - 0.00001875Y_1Y_2 + 0.000017708Y_2^2$
R_7	1.0000	$2.591431 - 0.04102X_1 + 0.000218X_2 - 0.009894Y_1 + 0.00429Y_2 + 0.000153X_1^2 + 0.00003125X_1X_2 - 0.0000375X_2^2 + 0.0000625X_1Y_1 + 0.00005625X_2Y_1 + 0.000015625Y_1^2 - 0.00001875X_1Y_2 - 0.00003125X_2Y_2 - 0.00001875Y_1Y_2 + 0.0000125Y_2^2$

Appendix C. Statistical analyses for random case via RSM

Tables 21 and 22

Table 21
F-ratio and important ranking for $R_1 \sim R_7$ (random case).

Factors	R ₁ (1)		R ₂ (2)		R ₃ (3)		R ₄ (4)		R ₅ (5)		R ₆ (6)		R ₇ (7)		Combined important ranking (1) + (2) + (3) + (4) + (5) + (6) + (7)
	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	F-Ratio	Rank	
X ₁	0.89	2	0.96	2	0.86	1	7.23	1	3.88	1	9.03	1	9.02	1	9
X ₂	0.91	1	0.43	3	0.59	2	0.23	4	3.50	2	0.55	3	0.55	3	16
Y ₁	0.10	4	1.40	1	0.23	4	1.18	2	0.67	3	1.06	2	1.06	2	18
Y ₂	0.12	3	0.24	4	0.25	3	0.68	3	0.62	4	0.43	4	0.43	4	25

Table 22
Response functions of $R_1 \sim R_7$ (random case).

Factor	R-Square	Response functions
R_1	0.4473	$81.014909 - 1.509146X_1 - 1.479351X_2 + 0.434993Y_1 - 0.14481Y_2 + 0.006952X_1^2 + 0.007492X_1X_2 + 0.008384X_2^2 + 0.000921X_1Y_1 + 0.000112X_2Y_1 - 0.003615Y_1^2 - 0.000284X_1Y_2 + 0.004004X_2Y_2 + 0.000069375Y_1Y_2 - 0.001973Y_2^2$
R_2	0.5614	$34.691304 - 0.727814X_1 - 0.388855X_2 + 0.101077Y_1 - 0.003472Y_2 + 0.003546X_1^2 + 0.002738X_1X_2 + 0.001657X_2^2 + 0.000587X_1Y_1 + 0.000070688X_2Y_1 - 0.001122Y_1^2 - 0.000016188X_1Y_2 + 0.000312X_2Y_2 + 0.000229Y_1Y_2 - 0.001134Y_2^2$
R_3	0.4498	$94.872366 - 2.242141X_1 - 2.121382X_2 + 1.315334Y_1 - 0.27839Y_2 + 0.010292X_1^2 + 0.010579X_1X_2 + 0.011484X_2^2 + 0.001899X_1Y_1 + 0.000382X_2Y_1 - 0.010331Y_1^2 - 0.000068125X_1Y_2 + 0.009907X_2Y_2 - 0.001455Y_1Y_2 - 0.005565Y_2^2$

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Table 22 (continued)

Factor	R-Square	Response functions
R ₄	0.7753	$15.980664 - 0.165519X_1 - 0.072286X_2 - 0.211677Y_1 + 0.037707Y_2 + 0.00031X_1^2 + 0.000334X_1X_2 + 0.000389X_2^2 + 0.00148X_1Y_1 + 0.00004375X_2Y_1 + 0.000653Y_1^2 - 0.000909X_1Y_2 + 0.000406X_2Y_2 + 0.00008125Y_1Y_2 + 0.00034Y_2^2$
R ₅	0.7577	$6.43678 - 0.03575X_1 - 0.037433X_2 - 0.115967Y_1 + 0.023702Y_2 - 0.000188X_1^2 - 0.00004675X_1X_2 + 0.00043X_2^2 + 0.001036X_1Y_1 - 0.000094563X_2Y_1 + 0.000284Y_1^2 - 0.000815X_1Y_2 + 0.000618X_2Y_2 + 0.000040813Y_1Y_2 + 0.000228Y_2^2$
R ₆	0.8086	$39.698567 - 0.480657X_1 - 0.150356X_2 - 0.472931Y_1 + 0.059838Y_2 + 0.001335X_1^2 + 0.001146X_1X_2 + 0.000398X_2^2 + 0.003059X_1Y_1 + 0.000295X_2Y_1 + 0.00153Y_1^2 - 0.001291X_1Y_2 + 0.000312X_2Y_2 + 0.000181Y_1Y_2 + 0.00062Y_2^2$
R ₇	0.8085	$33.035457 - 0.39996X_1 - 0.125123X_2 - 0.393563Y_1 + 0.049715Y_2 + 0.001111X_1^2 + 0.000954X_1X_2 + 0.000331X_2^2 + 0.002546X_1Y_1 + 0.000245X_2Y_1 + 0.001273Y_1^2 - 0.001074X_1Y_2 + 0.00026X_2Y_2 + 0.000151Y_1Y_2 + 0.000516Y_2^2$

Appendix D. MOO for different q values

Tables 23 and 24

Table 23

The optimal values under various q values (nonrandom case).

Variables	Optimal values										
	q										
	1	2	10	100	5000	10,000	20,000	50,000	100,000	200,000	∞
X ₁ [*]	79.2800	78.7916	79.0300	79.1842	78.575-5	75.9043	78.5755	78.5756	78.5756	78.5756	78.5756
X ₂ [*]	52.8300	50.4149	49.4382	49.5495	48.833-2	49.2659	48.8328	48.8328	48.8328	48.8328	48.8328
Y ₁ [*]	69.0060	73.006	73.006	69.8039	69.724-8	69.4454	69.721	69.7211	69.7211	69.7211	69.7211
Y ₂ [*]	9.0000	9.00003	9.0000	9.0000	9.7469	9.44785	9.746	9.7462	9.7462	9.7462	9.7462
Responses	Response values										
T	2.9529	1.5852	1.0183	0.9259	1.0099	1.7486	1.0098	1.0098	1.0098	1.0098	1.0098
R ₁	13.63%	25.14%	35.25%	36.42%	33.49%	9.41%	33.49%	33.48%	33.48%	33.48%	33.48%
R ₂	7.03%	6.65%	9.08%	11.29%	10.11%	4.82%	10.11%	10.11%	10.11%	10.11%	10.11%
R ₃	27.83%	35.25%	45.43%	46.36%	38.03%	12.11%	38.03%	38.03%	38.03%	38.03%	38.03%
R ₄	19.27%	19.38%	19.18%	19.17%	19.60%	21.43%	19.60%	19.60%	19.60%	19.60%	19.60%
R ₅	4.41%	10.39%	15.24%	15.81%	16.96%	6.95%	16.96%	16.96%	16.96%	16.96%	16.96%
R ₆	33.65%	33.56%	32.93%	32.80%	33.41%	37.27%	33.41%	33.41%	33.41%	33.41%	33.41%
R ₇	27.71%	27.64%	27.10%	27.00%	27.51%	30.74%	27.51%	27.51%	27.51%	27.51%	27.51%

Table 24

The optimal values under various q values (random case).

Variables	Optimal values										
	q										
	1	2	10	100	5000	10,000	20,000	50,000	100,000	200,000	∞
X ₁ [*]	78.1864	77.0301	78.5062	78.4700	77.681-3	77.7116	77.6640	77.6539	77.6483	77.6483	77.6483
X ₂ [*]	50.6260	50.7245	48.8300	48.8300	50.467-9	50.5595	50.3876	50.3886	50.3913	50.3913	50.3913
Y ₁ [*]	69.0060	73.006	73.006	73.006	69.944-8	70.2048	69.9254	69.9394	69.9434	69.9434	69.9434
Y ₂ [*]	9.0000	9.0000	13.0000	13.0000	9.2179	9.0184	9.1444	9.1458	9.1432	9.1432	9.1432
Responses	Response values										
T	3.2275	1.0189	0.4445	0.4123	0.6056	0.6166	0.6068	0.6075	0.6082	0.6082	0.6082
R ₁	13.64%	11.92%	12.76%	12.72%	14.06%	14.12%	14.06%	14.06%	14.06%	14.06%	14.06%
R ₂	5.77%	2.61%	4.48%	4.46%	5.39%	5.14%	5.36%	5.35%	5.35%	5.35%	5.35%
R ₃	15.23%	12.79%	12.04%	11.94%	17.60%	17.75%	17.62%	17.63%	17.62%	17.62%	17.62%
R ₄	19.70%	19.66%	19.51%	19.52%	19.73%	19.66%	19.75%	19.75%	19.76%	19.76%	19.76%
R ₅	8.87%	9.14%	8.36%	8.37%	8.93%	8.96%	8.94%	8.94%	8.95%	8.95%	8.95%
R ₆	34.34%	34.82%	33.90%	33.93%	34.46%	34.33%	34.48%	34.49%	34.49%	34.49%	34.49%
R ₇	28.75%	29.13%	28.38%	28.41%	28.85%	28.75%	28.87%	28.87%	28.87%	28.87%	28.87%

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